MODULAR APÉRY SYSTEMS: SOURCES, SLOPES, AND THE TWO-ROW DESIGN RULE

1. Introduction

An *Apéry system* for a real number ξ is a pair of sequences $a_n \in \mathbb{Z}$, $b_n \in \mathbb{Q}$ satisfying one linear recurrence with polynomial coefficients, with

$$a_n \asymp \lambda_1^n, \qquad |a_n\xi - b_n| \asymp \lambda_2^n, \qquad d_n^k b_n \in \mathbb{Z}, \qquad d_n = \mathrm{lcm}(1,\dots,n),$$

and ξ is irrational as soon as $\lambda_2 e^k < 1$. Apéry's systems for $\zeta(2)$ and $\zeta(3)$ satisfy this with room to spare; in the forty years since, a large number of structurally similar systems have been found—Zagier's and Almkvist–Zudilin's sporadic families, Cooper's, the Ramanujan-Machine continued fractions, and many constructed from modular forms—and none of them does. This paper is about why.

Beukers' modular interpretation [14] supplies the three ingredients of such a system: a genus-zero group Γ, a Hauptmodul t with integral q-expansion, and a modular form f of weight w with $\sum a_n t^n = f$. The companion b is then an Eichler integral, and the special value ξ is a period of the corresponding Eisenstein or cuspidal extension. Our starting point is that the three quantities in the inequality above are controlled by three *separate* pieces of this data: the special value by the *source* $\Phi = f \cdot Dt$ (Section 2); the rates λ_1, λ_2 by the position of the cusps and elliptic points of Γ in the coordinate t (Section 3); and the denominator depth k by the number of integrations minus the number that are free. A fourth quantity, invisible in the inequality but decisive for what follows, is the p-adic behaviour of the ratios b_n/a_n (Section 4).

Results. *Sources.* For the twelve sporadic families with modular parametrisations the source Φ is a holomorphic Eisenstein series with zero cuspidal projection (Theorem A, proved in the project papers cited there), and the Apéry limit is a rational multiple of a critical value $L(\Phi, w+1)$ (Theorem B). Eisenstein sources give Dirichlet L-values; cuspidal sources give critical values of cusp forms, of which we give two examples on the Domb curve, with limits $\frac{1}{2}L(f_6, 3)$ and $L(\eta(2\tau)^3\eta(6\tau)^3, 2)$. Period annihilation—the cancellation of unwanted critical values by oldform combinations, of which Apéry's $E_4 - 28E_4(2\tau) + 63E_4(3\tau) - 36E_4(6\tau)$ is the prototype—is finite Vandermonde algebra, and it produces new systems on demand (level-16 $\zeta(5)$, a level-12 $\zeta(7)$ parent with limit $\frac{209}{1728}\zeta(7)$, $\beta(4)$ at level 24, $L(4, \chi_{-3})$).

Slopes. For a three-term recurrence with constant term $c = \lambda_1\lambda_2$ the Casoratian is $-c^{n-1}/n^{w+1}$, so b_n/a_n converges p-adically with slope $\sigma_p = v_p(c) + 2\kappa_p$, where κ_p is the rate of p-power denominators of a_n (Proposition C). For integral rows the product formula gives $\sum_p \sigma_p \log p = \log|\lambda_1\lambda_2|$: archimedean imperfection is exactly redistributed as p-adic convergence. Apéry's rows ($c = \pm 1$) have none; Zagier's Catalan row ($c = 32$) has $\sigma_2 = 5$.

Rigidity. Rows sharing a period and a slope prime have p-adic limits in the same rational ratio as their archimedean limits (Conjecture D, verified in every available case, including one where the archimedean limit does not exist). This is the precise sense in which two realisations of one extension class are "aligned", and it is the mechanism behind the two-row constructions for Catalan's constant.

The two-row design rule. For two aligned rows the quality of the correlated-lattice construction of [4, 10] has a closed form (Theorem E) which reproduces every number in the literature (0.8579,

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0.9025, 0.6589, and the formally verified worthiness $1-\varepsilon$ of [12]) and a new one: the Domb and $(12,4,16)$ rows for $\zeta(3)$ are aligned at $p=2$ and give $\delta=0.9010$. At the balance point the product formula collapses the formula to a design rule, $\delta>1 \iff \log\Lambda_{\mathrm{dec}} > k\max(r,1)+r\log\rho_2^{\mathrm{eng}}$, which explains why exactly Catalan and $\zeta(3)$ come within ten percent of the threshold and every other known pair does not: the binding constraint is the existence of a *decaying* partner, and modular rows, being integral, never supply one—only hypergeometric rows with $\kappa_p>0$ do.

Past the threshold. The Catalan construction is supercritical: its 2-adic divisibility exceeds the break-even rate. We show that nothing expressible through valuations—extra congruences, factorisation of the cross determinant, extra rows, directional divisibility of coefficients—can exclude the rationality-kernel vector in that regime, and that the only remaining statement, Lemma P2 of Section 6, is of irrationality strength. This is a negative result with a positive reading: the method converts the irrationality of Catalan's constant, without loss, into a single statement about one explicit moment lattice, and we record what an attack on it must look like (Problem 7.5).

What this paper does not claim. No new irrationality theorem. Every displayed quality δ is either below 1 or, where the formal expression exceeds 1, accompanied by the control experiment showing that a rational number produces the same numbers. Evidence grades are attached to every statement: [PROVED] means a written proof or machine-checked certificate exists and is cited; *verified* means an exact finite computation to a stated range; conjectures are labelled as such. The computations are reproducible from the scripts listed in Appendix A.

Provenance. The results of Sections 2 and 3 and the Catalan constructions were obtained in an extended AI-assisted research programme (GPT-5, Claude) directed by the first author; the project papers cited as [3, 11, 4, 10, 13] contain the full proofs and certificates, and the Lean formalisation [12] of the $1-\varepsilon$ theorem depends only on the standard axioms. Sections 4–6 were obtained in the consolidation phase and are new.

2. Modular Apéry systems and their sources

This section fixes the objects, states the Source Theorem (Theorem A) and the limit theorem (Theorem B), and records the Hecke linear algebra that produces new sources on demand. Every statement carries its evidence class: [PROVED] means a proof exists and is cited to the place where it is written out; [VERIFIED: range] means an exact or high-precision finite computation over the stated range, which is never presented as a proof; [OPEN] is open.

2.1. Definitions.

Write $\theta_t = t\,d/dt$ and $\theta_q = q\,d/dq$, and let $d_n = \mathrm{lcm}(1,\dots,n)$.

Definition 2.1 (Apéry-like recurrence, sporadic families)**.** A recurrence of the shape

$$\text{(R2)}\qquad (n+1)^2A_{n+1} = (an^2+an+b)A_n - c\,n^2A_{n-1}, \tag{1}$$

$$\text{(R3)}\qquad (n+1)^3A_{n+1} = (2n+1)(an^2+an+b)A_n - n(cn^2+d)A_{n-1} \tag{2}$$

is *Apéry-like* when the solution with $A_0=1$, $A_1=b$ is an integer sequence. The known solutions form the *sporadic* list: six of type (R2) (Zagier's $\mathbf{A},\dots,\mathbf{F}$), six of type (R3) with $d=0$ (Almkvist–Zudilin's α (Domb), γ (Apéry's own), δ, ε, ζ, η), and Cooper's three further (R3) rows with $d\neq 0$ (s_7, s_{10}, s_{18}), which are of *operator* order two despite the cubic-looking recurrence [29, 1, 20]. Throughout, $r\in\{2,3\}$ denotes the order of the generating operator L (so $r=2$ for the nine second-order families, $r=3$ for the six third-order ones), and $w=r-1$ is the weight of the parametrizing form.

Definition 2.2 (nome, parametrizing form, rectification)**.** Let L be an order-r operator in θ_t over $\mathbb{Q}(t)$ with Frobenius basis $y_0 = 1+O(t)$, $y_1 = y_0\log t + g, \dots$ at $t=0$. The *nome* is $q = t\exp(g/y_0)$, so that $\log q = y_1/y_0$; write $t(q)$ for its inverse and $F(q) = y_0(t(q))$ for the *parametrizing form.* Say L is *exactly rectified* if

$$L = C(t) \cdot F \cdot \theta_q^r \cdot F^{-1} \tag{3}$$

for a rational function C; equivalently the kernel of L is $\operatorname{span}\{F, F\log q, \ldots, F(\log q)^{r-1}/(r-1)!\}$. For $r \geq 3$ this is strictly stronger than maximal unipotent monodromy [11, §1].

A *modular Apéry system* is a triple (Γ, t, F) in which t is a modular function on Γ with $t = q + O(q^2)$, F a holomorphic modular form of weight w with $F = 1 + O(q)$, and A_n is defined by $F = \sum_n A_n t^n$. Since $t \in q\mathbb{Z}[[q]]$ with leading coefficient 1, triangular peeling against the powers of t makes $A_n \in \mathbb{Z}$ automatic; integrality is therefore *not* the discriminating condition in this family, the recurrence is [23, §1].

Definition 2.3 (companion, source)**.** The *normalized companion* is the unique log-free power series y_B with $y_B(0) = 0$ and $L(y_B) = t$; equivalently $B_n = [t^n]y_B$ is the solution of the recurrence with $B_0 = 0$, $B_1 = 1$. The *companion source* is

$$\Phi := \frac{t}{C(t)F} = \frac{t\,\sigma^r}{P_r(t)\,F}, \qquad \sigma := \frac{\theta_q t}{t}, \tag{4}$$

where $P_2(t) = 1 - at + ct^2$ and $P_3(t) = 1 - 2at + ct^2$ are the leading symbols.

Theorem 2.4 (companion inversion; [11, Thm. 2.4])**.** [PROVED] *If L is exactly rectified as in* (3) *then*

$$y_B = F \cdot \theta_q^{-r}\Big(\frac{t}{C(t)F}\Big) = F\,\theta_q^{-r}\Phi,$$

where θ_q^{-1} is the constant-free antiderivative $\sum b_m q^m \mapsto \sum (b_m/m)q^m$. Consequently $B_n = [t^n]\,F\,\theta_q^{-r}\Phi$ for all $n \geq 0$.

Proof ([11, §2]). The displayed series is log-free and vanishes at $t = 0$, and $L(\cdot) = CF\theta_q^r(F^{-1}\cdot) = CF \cdot t/(CF) = t$ by (3). Two lemmas identify it with y_B: a boundary-defect computation shows $L(y_B) = t$ exactly (the coefficients of t^m, $m \neq 1$, are instances of the recurrence, and the coefficient of t is $1^r B_1 - (\cdots)B_0 = 1$), and a uniqueness statement shows that a log-free y with $y(0) = 0$ and $L(y) = t$ is unique, since two such differ by a log-free kernel element vanishing at 0 while the log-free kernel is $\mathbb{Q}y_0$ with $y_0(0) = 1$. □

Theorem 2.5 (rectification for the fifteen; [11, Thm. 3.3, Thm. 3.5])**.** [PROVED] *Every one of the fifteen sporadic operators is exactly rectified, with $C = P_r/\sigma^r$. For $r = 2$ this is Frobenius theory. For $r = 3$ each of the six operators equals $t^3P_3(t)$ times the symmetric square of the explicit second-order operator*

$$\widetilde{D} = \theta_t^2 - t\Big(2a\theta_t^2 + a\theta_t + \tfrac{b}{2}\Big) + c\,t^2\Big(\theta_t + \tfrac{1}{2}\Big)^2,$$

by exact operator identities in $\mathbb{Q}(t)$; if u_0, u_1 is a Frobenius basis of $\widetilde{D}$ then $F = u_0^2$, $u_1/u_0 = \log q$, and the kernel of L is $\{u_0^2, u_0u_1, u_1^2/2\} = \{F, F\log q, \frac{1}{2}F\log^2 q\}$.

Combining Theorems 2.4 and 2.5 gives, for all fifteen families and all $n \geq 0$,

$$B_n = [t^n]\;F \cdot \theta_q^{-r}\Phi, \qquad \Phi = \frac{t\sigma^r}{P_rF}. \tag{5}$$

In classical language, B is an *Eichler integral* of Φ: with $\Phi = \sum_{m\geq 1} c(m)q^m$,

$$\Theta := \theta_q^{-r}\Phi = \sum_{m\geq 1}\frac{c(m)}{m^r}\,q^m, \qquad y_B(t(q)) = F(q)\Theta(q). \tag{6}$$

The *denominator exponent* of the row is the least k with $d_n^k B_n \in \mathbb{Z}$ for all n; the r nominal integrations of (5) give $k \leq r$, and free integrations can lower it (§2.6).

2.2. **Theorem A: the Source Theorem.** The quotient (4) looks opaque. It is not: it collapses.

Proposition 2.6 (collapse of the source; [2, Prop. 2.1])**.** [PROVED] *For both operator families, in the canonical coordinate,*

$$\boxed{\Phi = F\,\theta_q t.} \tag{7}$$

Proof. Write $K = \theta_q \log t = \sigma$. For L_2, the Wronskian of F and $F \log q$ is $F^2\, d\log q/dt = 1/(tP_2(t))$, with the normalization fixed at $t = 0$; hence $K = P_2F^2$ and $\Phi = t(P_2F^2)^2/(P_2F) = tP_2F^3 = tKF = F\,\theta_q t$. For L_3, the Wronskian of the two solutions of $\widetilde{D}$ is $1/(t\sqrt{P_3})$; since $F = u_0^2$ this gives $K = F\sqrt{P_3}$, i.e. $K^2 = P_3F^2$, and $\Phi = tK^3/(P_3F) = tKF = F\,\theta_q t$. □

Since $\theta_q t$ is a meromorphic modular form of weight 2 and F has weight $w = r - 1$, (7) predicts $\operatorname{wt}\Phi = r + 1$. What is not predicted is the following.

Theorem A (Source Theorem). [PROVED] [2]. For each of the twelve modularly identifiable sporadic families

$$\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}, \quad \alpha, \gamma, \delta, \varepsilon, \zeta, \eta,$$

the companion source is $\Phi = F\,\theta_q t$, and it is exactly the holomorphic Eisenstein series of weight $r+1$ on the level of the family listed in Table 1. In particular the cuspidal projection of every one of these twelve sources is zero.

Structure of the proof ([2]). Proposition 2.6 identifies the source. For the eleven families whose (t, F) are ordinary eta quotients, Ligozat's criterion supplies the weight, character and cusp orders, and in particular the pole cancellation $\operatorname{ord}_{1/c} F = -\operatorname{ord}_{1/c} t$ at every cusp, so that $F\theta_q t = tFK$ is holomorphic everywhere. That the displayed eta quotients are the *canonical* parametrization — rather than forms whose first coefficients happen to agree — is proved by two small holomorphic identities, a normalized Wronskian identity ($K = P_2F^2$, resp. $K^2 = P_3F^2$) and a Rankin–Cohen form $H_2 = H_3 = 0$ of the differential equation, each verified by exact rational Fourier arithmetic through its Sturm bound (bounds ≤ 36; the largest are $\mathbf{B}$ and η). The source identity $F\theta_q t = E$ is then a third Sturm check in $M_{r+1}(\Gamma_0(N), \chi_F)$ (bounds ≤ 18). The remaining family $\mathbf{D}$ is treated on $\Gamma_1(5)$ using Zagier's Rogers–Ramanujan parametrization [29]: there $\Phi^2 = t\,\eta_1^9\eta_5^3$ is a holomorphic weight-six form, $E_{\mathbf{D}}^2$ lies in the same space, and the weight-six Sturm bound 12 on $\Gamma_1(5)$ converts a finite coefficient check into the identity $\Phi^2 = E_{\mathbf{D}}^2$; both series begin with q, so $\Phi = E_{\mathbf{D}}$. □

Three remarks on Table 1. First, the normalization $\Phi = q + O(q^2)$ is forced by $t = q + O(q^2)$, $F = 1 + O(q)$, $P = 1 + O(t)$; the constant terms of the Eisenstein combinations therefore cancel identically, which they do in every family, a check not part of the fitted data. Second, the γ row is, up to normalization, $\frac{1}{240}(-E_4(q) + 28E_4(q^2) - 63E_4(q^3) + 36E_4(q^6))$, exactly Beukers' weight-four level-six series for Apéry's $\zeta(3)$ case [14], here re-derived from the recurrence with no classical input; $\mathbf{D}$ likewise reproduces the $\Gamma_1(5)$ weight-three picture for $\zeta(2)$. These two are controls. Third, ζ is the cleanest object in the table: the primitive (χ_{-3}, χ_{-3})-Eisenstein newform of weight 4 and level 9, with coefficient exactly 1.

Remark 2.7 (the L-column). The right-hand column of Table 1 is the Dirichlet-series factorization of a divisor convolution: if $c(m) = \sum_{d|m} \chi_1(d)\chi_2(m/d)d^k$ then $\sum_m c(m)m^{-s} = L(\chi_1, s-k)L(\chi_2, s)$, and the oldform coefficients contribute the finite Euler-type factor $P_C(s) = \sum_d c_d d^{-s}$ (§2.4). For α and δ the finite factors are $P_\alpha(s) = 1 - 17\cdot 2^{-s} - 9\cdot 3^{-s} + 16\cdot 4^{-s} + 153\cdot 6^{-s} - 144\cdot 12^{-s}$ and $P_\delta(s) = 1 - 14\cdot 2^{-s} - 3^{-s} + 16\cdot 4^{-s} + 14\cdot 6^{-s} - 16\cdot 12^{-s}$. This is the conceptual answer to why the sporadic limit column consists of zeta and Dirichlet L-values: because the sources are Eisenstein.

TABLE 1. The twelve companion sources $\Phi = \sum_{m\geq 1} c(m)q^m$, and the Dirichlet series of the associated L-function. Here $\sigma_\chi^{(k)}(m) = \sum_{d|m}\chi(m/d)d^k$ (inner placement), $\widetilde{\sigma}_\chi^{(k)}(m) = \sum_{d|m}\chi(d)d^k$ (outer), $\sigma_3(m) = \sum_{d|m} d^3$, and $\sigma(m/k) := 0$ unless $k \mid m$; ψ_1, ψ_2 are the real and imaginary parts of the quartic character mod 5 normalized by $\chi(2) = i$. Weight is $r+1$. [PROVED] (Theorem A).

fam	r	N	$c(m)$	$L(\Phi, s)$
A	2	6	$\widetilde{\sigma}_{\chi_{-3}}^{(2)}(m) - \widetilde{\sigma}_{\chi_{-3}}^{(2)}(m/2)$	$(1-2^{-s})\zeta(s)L(\chi_{-3}, s-2)$
B	2	36	$\sigma_{\chi_{-3}}^{(2)}(m) - 6\sigma_{\chi_{-3}}^{(2)}(m/2) - 8\sigma_{\chi_{-3}}^{(2)}(m/4)$	$(1-6\cdot 2^{-s}-8\cdot 4^{-s})L(\chi_{-3}, s)\zeta(s-2)$
C	2	6	$\sigma_{\chi_{-3}}^{(2)}(m) - 8\sigma_{\chi_{-3}}^{(2)}(m/2)$	$(1-8\cdot 2^{-s})L(\chi_{-3}, s)\zeta(s-2)$
D	2	5	$\sum_{d\mid m}(\psi_1(d) - 2\psi_2(d))d^2$	$(\frac{1}{2}+i)\zeta(s)L(\chi, s-2) + \text{c.c.}$
E	2	8	$\sigma_{\chi_{-4}}^{(2)}(m) - 8\sigma_{\chi_{-4}}^{(2)}(m/2)$	$(1-8\cdot 2^{-s})\beta(s)\zeta(s-2)$
F	2	12	$\sigma_{\chi_{-3}}^{(2)}(m) - 7\sigma_{\chi_{-3}}^{(2)}(m/2) - 8\sigma_{\chi_{-3}}^{(2)}(m/4)$	$(1-7\cdot 2^{-s}-8\cdot 4^{-s})L(\chi_{-3}, s)\zeta(s-2)$
α	3	12	$\sigma_3(m) - 17\sigma_3(m/2) - 9\sigma_3(m/3) +$ $16\sigma_3(m/4) + 153\sigma_3(m/6) - 144\sigma_3(m/12)$	$P_\alpha(s)\,\zeta(s)\zeta(s-3)$
γ	3	6	$\sigma_3(m) - 28\sigma_3(m/2) + 63\sigma_3(m/3) -$ $36\sigma_3(m/6)$	$(1-28\cdot 2^{-s}+63\cdot 3^{-s}-36\cdot 6^{-s})\zeta(s)\zeta(s-3)$
δ	3	12	$\sigma_3(m) - 14\sigma_3(m/2) - \sigma_3(m/3) +$ $16\sigma_3(m/4) + 14\sigma_3(m/6) - 16\sigma_3(m/12)$	$P_\delta(s)\,\zeta(s)\zeta(s-3)$
ε	3	8	$\sigma_3(m) - 21\sigma_3(m/2) + 84\sigma_3(m/4) -$ $64\sigma_3(m/8)$	$(1-2^{-s})(1-4\cdot 2^{-s})(1-16\cdot 2^{-s})\zeta(s)\zeta(s-3)$
ζ	3	9	$\sum_{d\mid m}\chi_{-3}(d)\chi_{-3}(m/d)d^3$	$L(\chi_{-3}, s)L(\chi_{-3}, s-3)$
η	3	20	$\sigma_{\chi_5}^{(3)}(m) - 14\sigma_{\chi_5}^{(3)}(m/2) - 16\sigma_{\chi_5}^{(3)}(m/4)$	$(1-14\cdot 2^{-s}-16\cdot 4^{-s})L(\chi_5, s)\zeta(s-3)$

Remark 2.8 (Cooper's three; scope). Theorem A asserts nothing about s_7, s_{10}, s_{18}. Their canonical nomes are integral ($t \in q\mathbb{Z}[[q]]$, no rescaling), but their (t, F) are not eta quotients in that normalization, and no weight-3 or weight-4 Eisenstein fit exists at levels $\{7, 10, 18\}$ and their divisors with the natural characters [EXCLUDED: stated bases]. The obstruction is the coordinate, not integrality; it would be wrong to say Cooper's three are not Eisenstein. Finding the Atkin–Lehner normalization or hauptmodul in which they are eta-type is [OPEN] [3, Prob. Φ-2].

Remark 2.9 (novelty scope). The second-order side overlaps substantially with Zagier's period analysis: weight-three Eisenstein series and their Eichler integrals already occur explicitly there for the six (R2) rows [29], and the γ row is Beukers' [14]. No discovery of Eisenstein behaviour is claimed for those individual families. What is new in [2] is the uniform identification of the canonical companion source as $F\theta_q t$, the twelve-family source table in one normalization, and a single finite certification that simultaneously proves canonicality of the eta parametrizations.

2.3. **Theorem B: the Apéry limit is a critical value.** Let t_c be the root of P_r of smallest modulus and q_c its nome. In the q-coordinate the map $q \mapsto t(q)$ *folds* at q_c: $dt/dq = 0$ there and $t - t_c \simeq \kappa(q-q_c)^2$ with $\kappa \neq 0$. This is the modular reason for the square-root singularity of t.

Lemma 2.10 (fold connection lemma; [3, Lem. 4.1]). [PROVED] *Let $t(q)$ be analytic near q_c with a simple fold there, and let $y_0 \circ t = F$ and $y_B \circ t = F\Theta$ be analytic at q_c. Then the singular parts of y_0, y_B at t_c are their odd parts under the local fold involution, and, provided $F'(q_c) \neq 0$,*

$$\xi := \lim_{n\to\infty}\frac{B_n}{A_n} = \frac{(y_B\circ t)'(q_c)}{(y_0\circ t)'(q_c)} = \Theta(q_c) + \frac{F(q_c)\,\Theta'(q_c)}{F'(q_c)}. \tag{8}$$

Proof. Locally at q_c put $u = q - q_c$. The involution ι with $t(\iota q) = t(q)$ is analytic, fixes q_c, and satisfies $\iota(u) = -u + O(u^2)$; linearizing it (Koenigs) we may take $\iota(v) = -v$ and $t - t_c = \kappa v^2$. A function h analytic at q_c splits as $h^{\mathrm{ev}} + h^{\mathrm{odd}}$ with h^{ev} an analytic function of v^2, hence of t — the part regular at t_c — and h^{odd} equal to $\sqrt{t - t_c}$ times an analytic function of t — the square-root singular part. If $y = g + \sqrt{1 - t/t_c}\, h$ near t_c with g, h analytic and $h(t_c) \neq 0$, singularity analysis gives $[t^n]y \sim -h(t_c)t_c^{-n}n^{-3/2}/(2\sqrt{\pi})$, the analytic part contributing $O(\rho^{-n})$ with $\rho > |t_c|$. The same prefactor occurs for y_0 and y_B, so $\xi = h_B(t_c)/h_0(t_c)$, which equals the ratio of full q-derivatives at q_c because the even parts have vanishing v-derivative there and dv/dq cancels. Substituting $y_0 \circ t = F$, $y_B \circ t = F\Theta$ gives (8). □

Two hypotheses do work here and are flagged: $F'(q_c) \neq 0$ (visible from the eta formulas in every family), and analyticity of Θ at q_c, which for $|q_c| < 1$ follows from $c(m) = O(m^{r+\epsilon})$.

Theorem B (Apéry limit = critical value). Let a modular Apéry system be exactly rectified with source Φ of weight $r+1$ and a simple fold at a real q_c with $F'(q_c) \neq 0$. Then the limit $\xi = \lim B_n/A_n$ exists and is given by (8).

(i) *Eisenstein source.* For the twelve families of Theorem A, Θ is the Eichler integral of an Eisenstein series whose Dirichlet series factors as in Table 1, and ξ is a rational multiple of a critical value of that factorization: $\zeta(3)$ for the trivial-character weight-four rows $\gamma, \alpha, \varepsilon$; $\zeta(2)$ for the trivial-character weight-three rows $\mathbf{A}, \mathbf{D}$; $L(\chi_{-3}, 2)$ and $L(\chi_{-3}, 3)$ for the χ_{-3}-twisted rows $\mathbf{C}, \mathbf{F}, \zeta$; Catalan's $G = L(\chi_{-4}, 2)$ for $\mathbf{E}$.

(ii) *Cuspidal source.* For a source lying in S_{r+1} and odd for the Fricke involution whose fixed point is the fold, ξ is a rational multiple of $L(f, r)$, with no quasiperiod contribution.

Part (i) is proved for the nine families with a real fold in the strong sense that both the source (Theorem A) and the connection formula (Lemma 2.10) are proved; what is *not* written in general is the closed evaluation of (8) producing the rational prefactor $(1/6, 7/32, 5/8, \dots)$ uniformly. The nine limits have been recomputed from the identified source alone [VERIFIED: numeric, series order $N = 26$ with truncation tracked as $|q_c|^N$]: $\gamma \to \zeta(3)/6$ to $6.6 \cdot 10^{-15}$, $\varepsilon \to 7\zeta(3)/32$ to $8.8 \cdot 10^{-13}$, $\alpha \to 7\zeta(3)/24$ to $8.3 \cdot 10^{-8}$, $\mathbf{D} \to \zeta(2)/5$ to $3.7 \cdot 10^{-7}$, and $\zeta, \mathbf{A}, \mathbf{C}, \mathbf{E}, \mathbf{F}$ to between $4.7 \cdot 10^{-4}$ and $3.6 \cdot 10^{-2}$, the last five having $|q_c|$ large enough that $|q_c|^{26}$ is itself of that order [3, Tab. 2]. The right-hand column measures the geometry at that series order, not the theory.

Remark 2.11 (the three complex folds). [PROVED] $\mathbf{B}, \delta, \eta$ are exactly the sporadic families whose $P(t)$ has a complex-conjugate pair of roots, with disc $= -27, -128, -16$ and splitting fields $\mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(i)$. Two dominant singularities of equal modulus and non-real ratio make B_n/A_n spiral; that is all "no limit" means. The connection value (8) is still defined at complex q_c, and the resulting canonical constants are recorded to 15, 39 and 33 digits [VERIFIED: numeric, $N = 60$, dps 80]; no rational relation was found for them over natural L-value bases [EXCLUDED: stated basis, max coefficient 10^6–10^7]. Recognizing ξ_δ in closed form is [OPEN] [3, Prob. Φ-1].

The cuspidal case: the Domb apparatus. Part (ii) of Theorem B is not vacuous: it is realized, on Domb's curve, by a deliberately manufactured cuspidal source [8].

Proposition 2.12 (the fold is the Fricke point; [8, Prop. 5.1])**.** [PROVED] *For the family α (Domb), $P_\alpha(t) = 1 - 20t + 64t^2 = (1 - 4t)(1 - 16t)$, level* 12*, the dominant fold is $\tau_{12} = i/\sqrt{12}$, with $t_c = 1/16$ and conjugate root $t'_c = 1/4$.*

Since $\dim S_4(\Gamma_0(12)) = 2$ with no newform, the space is the oldspace of the level-6 newform $f_6 = (\eta_1\eta_2\eta_3\eta_6)^2$, and the Fricke involution splits it.

Theorem 2.13 (oldspace swap and vanishing; [8, Thm. 5.3])**.** [PROVED] $f_6|_4W_6 = f_6$ *(directly from $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$), and on $S_4(\Gamma_0(12)) = \mathrm{span}\{f_6(q), f_6(q^2)\}$ the involution W_{12} acts by*

$f_6(q) \mapsto 4f_6(q^2)$, $f_6(q^2) \mapsto \frac{1}{4}f_6(q)$. Hence the eigenvectors are $f_6 \mp 4f_6(q^2)$, and the odd one

$$f^* := f_6(q) - 4f_6(q^2) = q - 6q^2 - 3q^3 + 12q^4 + 6q^5 + \cdots$$

satisfies $f^*|_4W_{12} = -f^*$ and $f^*(\tau_{12}) = 0$.

Proof. The level-12 slash is $(f_6|_4W_{12})(\tau) = f_6(-1/(6\cdot 2\tau))(\sqrt{12}\tau)^{-4} = (\sqrt{6}\cdot 2\tau)^4 f_6(2\tau)(\sqrt{12}\tau)^{-4} = 4f_6(2\tau)$, using $\varepsilon(f_6, W_6) = +1$ and $(24/12)^2 = 4$; the second formula follows since W_{12} is an involution. At the fixed point $-1/(12\tau_{12}) = \tau_{12}$ the weight-four automorphy factor is $(\sqrt{12}\tau_{12})^4 = i^4 = 1$, so $f^*(\tau_{12}) = -f^*(\tau_{12})$. □

More generally: if $g \in M_k(\Gamma_0(N))$ with $g|_kW_N = -g$ and $k \equiv 0 \pmod 4$, then g vanishes at $i/\sqrt{N}$. The mechanism now runs on one parity, through two independent channels.

Theorem 2.14 (the level-12 apparatus; [8, §5–§6])**.** [PROVED] *With the branch $\sqrt{1-4t} = 1+O(q)$ one has the exact identity $f^*\sqrt{1-4t} = \Phi_\alpha$, hence $R^*(t) = tf^*/\Phi_\alpha = t/\sqrt{1-4t} = \sum_{m\geq 1}\binom{2m-2}{m-1}t^m$. Consequently the f^*-companion B^* is defined over $\mathbb{Z}$ by*

$$m^3B^*_m = (2m-1)\big(10(m-1)^2+10(m-1)+4\big)B^*_{m-1} - 64(m-1)^3B^*_{m-2} + \binom{2m-2}{m-1}, \quad B^*_0 = 0,\ B^*_1 = 1,$$

and satisfies $d_n^3B^*_n \in \mathbb{Z}$ for every n, $\limsup_n |B^*_n/A_n - \xi^*|^{1/n} \leq t_c/t'_c = \frac{1}{4}$, and

$$\boxed{\xi^* = \lim_{n\to\infty}\frac{B^*_n}{A_n} = L(f^*,3) = \frac{1}{2}L(f_6,3).}$$

The two channels are worth naming. *Fold regularization*: R^* is analytic on $|t| < 1/4$, so the inhomogeneity's singularity has moved off the fold onto the conjugate root and the error ratio becomes the root ratio t_c/t'_c. *Period parity*: with $\Lambda_*(s) = 12^{s/2}(2\pi)^{-s}\Gamma(s)L(f^*,s)$ one has $\Lambda_*(s) = -\Lambda_*(4-s)$, so $L(f^*,2) = 0$ and only the odd critical datum survives; the rational factor $\frac{1}{2}$ then comes from $L(f^*,s) = (1-4\cdot 2^{-s})L(f_6,s)$ and from differentiating the exact Eichler cocycle at τ_{12}. By contrast, on Apéry's own curve at level 6 the Fricke sign is $+1$, $f_6(\tau_c) \neq 0$, and the fold value is contaminated: $\Theta(\tau_c) = L(f_6,3) - \frac{\pi}{\sqrt{6}}L(f_6,2)$. Fricke-oddness is thus the single operative hypothesis, and a classification corollary in [8] shows α is the unique order-three sporadic family admitting the construction.

A weight-three cuspidal row. The same phenomenon occurs one weight lower, and there it is cheaper. A mechanical scan of eta-quotient pairs (t, F) over all fourteen genus-zero levels (520 005 candidates) reproduces all six third-order and five of the six second-order sporadic rows and produces, over the Domb parameter $t = -(\eta_2\eta_6/\eta_1\eta_3)^6$ with the *weight-one* form $F = \eta_1^2\eta_3^2/(\eta_2\eta_6)$ (so that F^2 is the Domb weight-two form: this row is the Sym^1 to Domb's Sym^2), the sequence $a_n = 1, 2, 12, 104, 1078, 12348, \ldots$ satisfying

$$(n+1)^2a_{n+1} = (20n^2+10n+2)\,a_n - 16(2n-1)^2\,a_{n-1} \tag{9}$$

[VERIFIED: q-expansion, every $n \leq 208$] [23]. This is not a sixteenth sporadic row: (9) is outside Zagier's symmetric normalization, since p_1 is not symmetric under $n \mapsto -n-1$ and $p_0 = 16(2n-1)^2 \neq cn^2$. Its companion has $d_n^2b_n \in \mathbb{Z}$ [VERIFIED: $n \leq 200$], i.e. $k = 2$, and

$$\lim_{n\to\infty}\frac{b_n}{a_n} = L(f,2), \qquad f = \eta(2\tau)^3\eta(6\tau)^3 \in S_3(\Gamma_0(12), \chi_{-3}), \tag{10}$$

the weight-three CM newform of level 12, $f = q - 3q^3 + 2q^7 + 9q^9 - 22q^{13} + \cdots$ [VERIFIED: 60 digits; `lindep` relation exactly $[-1,0,1]$, so the limit is $L(f,2)$ on the nose with no rational factor, and the eigenform was produced independently as `mfinit([12,3,-3],0)` and as `mffrometaquo([2,3;6,3])`]. Its characteristic roots are 16 and 4, as Domb's, so the rate is again 4^{-n}, but the denominator exponent is 2 rather than 3. The comparison is the point:

row	source weight	limit	k	$\log\Lambda - k$
Domb apparatus (Thm. 2.14)	4 (f_6, level 6)	$L(f_6,3)/2$	3	-0.2274
weight-three row (9)	3 (f, level 12, χ_{-3})	$L(f,2)$	2	$+0.7726$

That single unit of denominator flips the sign of the budget of §5.3, and $+0.7726$ is the largest in the census. It is a ceiling, not an achievement: this row has $|\lambda_2| = 4 > 1$, so its own linear form grows and it proves nothing alone. Its use is as an *engine* (§5.4). **[TODO: The identification (10) is [VERIFIED: 60 digits] only; the analogue of Theorem 2.14 — a proved source identity $F\theta_q t =$ (explicit form) and a Fricke-parity argument at weight three — has not been written. It should be, since the weight-three CM setting is if anything simpler than the level-12 weight-four one.]**

2.4. **Period annihilation as Hecke linear algebra.** Theorem A explains the existing table; the following linear algebra manufactures new rows. For $d \geq 1$ write $(V_d f)(\tau) = f(d\tau)$. A finite oldform correspondence $C = \sum_{d\in\mathcal{D}} c_d V_d$ has Mellin eigenvalue $P_C(s) = \sum_d c_d d^{-s}$, whence

$$L(C\Phi, s) = P_C(s)\, L(\Phi, s): \tag{11}$$

zeros of P_C annihilate selected critical and endpoint components. This is the finite shadow of the extension-class data; Franke's exact theory of Eisenstein spaces with prescribed vanishing critical L-values in a nonprincipal-character sector is substantially broader, and the projectors below are Hecke-generated spectral slices of Franke-type critical fibres [26].

Theorem 2.15 (barycentric period interpolation)**.** [PROVED] *Let $d_1, \dots, d_m$ be distinct positive rationals, $P(s) = \sum_i c_i d_i^{-s}$, and fix $\alpha \in \mathbb{Z}$, $q \geq 1$. Impose $P(\alpha + 2j) = 0$ for $j = 0, \dots, q-1$. Put $x_i = d_i^{-2}$ and $Q(x) = \prod_i (x - x_i)$. Then the complete solution space is*

$$\boxed{c_i = d_i^{\alpha}\frac{R(x_i)}{Q'(x_i)}, \qquad \deg R \leq m - q - 1,}$$

so the nullspace has dimension $m - q$ when $m \geq q + 1$.

Proof. Writing $w_i = c_i d_i^{-\alpha}$ the conditions become $\sum_i w_i x_i^j = 0$ for $0 \leq j < q$. For $\deg H \leq m-2$ the partial-fraction identity $\sum_i H(x_i)/Q'(x_i) = 0$ holds; apply it with $H(x) = R(x)x^j$. The dimension count agrees with the Vandermonde nullity, so the family is exhaustive. □

Proposition 2.16 (Mellin-degree shift)**.** [PROVED] *Since $DV_d = dV_d D$ for $D = \theta_q$, one has $D^a V_d D^{-a} = d^a V_d$ wherever defined on the zero-constant sector, hence $D^a C D^{-a} = \sum_d c_d d^a V_d$ and $P_{D^a C D^{-a}}(s) = P_C(s-a)$: differential conjugation shifts every prescribed Mellin zero by a. On the depth-r Eichler primitive the same relation gives $D^{-r}V_d = d^{-r}V_d D^{-r}$, so one operation has three compatible realizations — source $\sum c_d V_d$, normal function $\sum c_d d^{-r} V_d$, target regulator $P_C(r)$.*

For a target $L(r,\chi)$ of opposite parity the lower inner-orientation nuisance slots are $J_r = \{j : 1 \leq j < r,\ r - j \text{ odd}\}$, so the minimal critical projector at one prime p is $C^{\mathrm{crit}}_{r,p} = \prod_{j\in J_r}(1 - p^j V_p)$. For trivial character and odd $r = 2m+1$ the two orientations coincide, and full endpoint/critical purification gives

$$C^{\mathrm{full}}_{r,p} = \prod_{a=0}^{m+1}\left(1 - p^{2a}V_p\right) = \sum_{j=0}^{m+2}(-1)^j p^{j(j-1)}\begin{bmatrix} m+2 \\ j\end{bmatrix}_{p^2} V_p^{\,j} \tag{12}$$

by the q-binomial theorem, so the coefficients are Gaussian binomials.

Example 2.17 ($\zeta(3)$ and $\zeta(5)$)**.** For $r = 3$, $p = 2$: $C^{\mathrm{full}}_{3,2} = (1 - V_2)(1 - 4V_2)(1 - 16V_2) = 1 - 21V_2 + 84V_4 - 64V_8$, which is exactly the level-8 ε source of Table 1. For $r = 5$, $p = 2$: $C^{\mathrm{full}}_{5,2} = 1 - 85V_2 + 1428V_4 - 5440V_8 + 4096V_{16}$, the level-16 purified weight-six $\zeta(5)$ source found independently in the project calculations. Thus $\zeta(5)$ is the first odd-zeta case whose raw critical completion is generically at least two-dimensional.

For trivial character and even weight k the reflected positions s and $k-s$ are exchanged by Fricke, so in a fixed eigenspace only one condition per reflected pair survives.

Theorem 2.18 (sign-refined Fricke count)**.** [PROVED] *Let k be even, $\epsilon = \pm 1$. The number of independent even-period conditions in the W_N-eigenspace of sign ϵ is*

$$q_\epsilon(k) = \begin{cases} (k+2)/4, & k \equiv 2 \pmod 4, \\ k/4+1, & k \equiv 0 \pmod 4),\ \epsilon = +1, \\ k/4, & k \equiv 0 \pmod 4),\ \epsilon = -1, \end{cases}$$

and if $h_\epsilon(N)$ is the dimension of the divisor-oldform Fricke eigenspace, the purified source dimension is $\max\{0, h_\epsilon(N) - q_\epsilon(k)\}$.

Proof sketch ([6, §5]). Fricke symmetry relates $P(s)$ and $P(k-s)$, so reflected conditions are equivalent; when $k \equiv 0 \pmod 4$ the central point $s = k/2$ is forced to vanish in the anti-Fricke sector. Pairing divisors d and N/d and setting $z_d = d^2/N + N/d^2$ reduces the remaining matrix, after nonzero scalings, to a Chebyshev/Vandermonde evaluation matrix in distinct z_d. □

Proposition 2.19 (paired-character count, $r = 4$, $\chi = \chi_{-4}$)**.** [PROVED] *In the divisor-oldform sector at level $4D$ paired by W_{4D}, each $\pm i$ Fricke eigenspace of fully purified $\beta(4)$ sources has dimension* $\max\{0, \tau(D) - 3\}$.

Proof. After Fricke pairing the three inner conditions are evaluations at the arithmetic progression $1, 3, 5$; Theorem 2.15 gives rank 3 on distinct divisor nodes, hence nullity $\tau(D) - 3$ when nonnegative. □

Four examples. (a) Level-16 $\zeta(5)$. Example 2.17: the minimal same-prime full projector (12) at $r = 5$, $p = 2$ is Fricke-even, while the successful level-12 scalar $\zeta(5)$ construction lies in the anti-Fricke sector and uses an independent weight-four companion. Theorem 2.18 predicts the uniqueness of that source in its divisor-oldform eigenspace.

(b) The hidden $\zeta(7)$ parent, level 12. The sign-refined classification exposes a fully purified anti-Fricke level-12 weight-eight source

$$\Phi_{12}^- = \tfrac{1}{480}\big(E_8 - 572E_8(2\tau) + 11583E_8(3\tau) - 36608E_8(4\tau) + 46332E_8(6\tau) - 20736E_8(12\tau)\big), \qquad L(\Phi_{12}^-, 7) = \frac{209}{1728}\zeta(7)$$

Its $p = 2$ completion $(1 - 16V_2)\Phi_{12}^-$ recovers the previously known level-24 source, and its $p = 3$ completion gives a level-36 scalar system with target $\frac{2717}{23328}\zeta(7)$. The classification thus found a hidden parent and explained an existing higher-level object as an isogeny completion.

(c) $\beta(4)$ at level 24. Take $r = 4$, $\chi = \chi_{-4}$, with the orientation doublet $S = E_{5,\chi_{-4},1}$, $T = E_{5,1,\chi_{-4}}$, $L(S, s) = \beta(s)\zeta(s-4)$, $L(T, s) = \zeta(s)\beta(s-4)$. On divisor support $(1, 2, 3, 6)$ put

$$\Phi_{\text{in}} = S - 56S(2\tau) + 189S(3\tau) - 216S(6\tau), \qquad \Phi_{\text{out}} = T - 28T(2\tau) + 63T(3\tau) - 36T(6\tau).$$

Then $P_{\text{in}}(1) = P_{\text{in}}(3) = P_{\text{in}}(5) = 0$, $P_{\text{out}}(0) = P_{\text{out}}(2) = P_{\text{out}}(4) = 0$, and $P_{\text{in}}(4) = -\frac{1}{3}$; since $\zeta(0) = -\frac{1}{2}$,

$$L(\Phi_{\text{in}}, 4) = \tfrac{1}{6}\beta(4). \tag{13}$$

By Proposition 2.19, at level 16 ($D = 4$) the obstruction determinant is $\det\left(\begin{smallmatrix} 1 & 1/2 & 1/4 \\ 1 & 1/8 & 1/64 \\ 1 & 1/32 & 1/1024 \end{smallmatrix}\right) = -\frac{135}{8192} \neq 0$, so no fully purified Fricke eigensource exists; at level 24 ($D = 6$) the eigenspace is one-dimensional with normalized inner vector $(1, -56, 189, -216)$. Level 24 is therefore the first divisor-oldform Fricke realization in this sector. The outer vector is exactly the Beukers annihilator $C_B = 1 - 28V_2 + 63V_3 - 36V_6$ and the inner one is its differential conjugate DC_BD^{-1} (Proposition 2.16): the two character orientations are adjacent Mellin-degree realizations of one Hecke correspondence. The realization has $A_n \in \mathbb{Z}$, $4d_n^4 b_n \in \mathbb{Z}$ and $b_n/a_n \to \beta(4)/6$; the rational trace coordinate forgets a degree-four modular cover, so the natural rational Picard–Fuchs module has rank $4 \cdot 4 = 16$, which is what forces the direct-image refinement of Layer A.

(d) $L(4, \chi_{-3})$. The canonical source of Shvets's level-three signature-three symmetric cube is $E_{5,\chi_{-3},1}$ with $L(E_{5,\chi_{-3},1}, 4) = -\frac{1}{2}L(4, \chi_{-3})$ [28]. The classification manufactures a fully purified level-24 $\pm i$ Fricke doublet with target $\frac{7}{32}L(4, \chi_{-3})$ and a rational Galois-trace Apéry pair with $A_n \in \mathbb{Z}$, $d_n^4 B_n \in \mathbb{Z}$, $B_n/A_n \to \frac{7}{32}L(4, \chi_{-3})$.

(e) $L(2, \chi_{-3})$ *and CDT.* Calegari–Dimitrov–Tang prove the $\mathbb{Q}$-linear independence of $1, \zeta(2), L(2, \chi_{-3})$ by arithmetic holonomy [19]. Their two weight-three source forms are exactly the inner and outer orientations predicted here, $(1 - 8V_2)E_{3,\chi_{-3},1}$ and $(1 - V_2)E_{3,1,\chi_{-3}}$; the hidden fully purified source is $(1 - 2V_2)(1 - 8V_2)E_{3,\chi_{-3},1} = 3\Phi_{\mathbf{C}} - 2\Phi_{\mathbf{F}}$. Their argument deliberately retains both a non-Tate and a Tate period coordinate, which is a realization-level lesson: full purification is not universally optimal.

2.5. **Isogeny, CM values, and a Lambert identity.** Isogeny acts on the rank-two local system and functorially on symmetric powers, so it transports the homogeneous skeleton without determining the extension class. Concretely, a level-6 parent $\zeta(3)$ source admits prime completions $\Phi_p = (1 - p^2V_p)\Phi_{\text{parent}}$; for $p = 5, 7, 11$ these land on genus-zero elliptic Apéry systems with exact targets $\frac{7}{20}\zeta(3)$, $\frac{3}{8}\zeta(3)$, $\frac{35}{88}\zeta(3)$.

When the fold sits at an Atkin–Lehner fixed point the Eichler cocycle collapses to a pure CM evaluation. For $\beta(4)$ at level 24, with the Fourier normalizations of (c) above, $\Phi_{\text{in}}|W_{24} = \frac{i\sqrt{6}}{16}\Phi_{\text{out}}$ and $\Phi_{\text{out}}|W_{24} = \frac{8i\sqrt{6}}{3}\Phi_{\text{in}}$, whose exchange constants multiply to -1; hence $\Phi_\pm = \Phi_{\text{in}} \pm \frac{\sqrt{6}}{16}\Phi_{\text{out}}$ satisfy $\Phi_\pm|W_{24} = \pm i\Phi_\pm$. With $\Theta_+ = D^{-4}\Phi_+$ and $C_4 = \frac{1}{6}\beta(4)$, the six unwanted slots having been killed, the transformation collapses to $\Theta_+(W_{24}\tau) - C_4 = i(\sqrt{24}\tau)^{-3}(\Theta_+(\tau) - C_4)$, whose multiplier at $\tau_* = i/\sqrt{24}$ is -1; therefore

$$\Theta_+(\tau_*) = \tfrac{1}{6}\beta(4), \tag{14}$$

with no separate π^4 correction. Expanding (14) in Lambert series gives, with $y = \pi/\sqrt{6}$,

$$J_4(y) = \sum_{n\geq 1} \frac{\chi_{-4}(n)}{n^4(e^{ny} - 1)}, \qquad K_4(y) = \sum_{n\geq 1} \frac{1}{n^4 \cosh(ny)},$$

the identity

$$\beta(4) = 6J_4(y) - 21J_4(2y) + 14J_4(3y) - J_4(6y) + \frac{3\sqrt{6}}{16}K_4(y) - \frac{21\sqrt{6}}{64}K_4(2y) + \frac{7\sqrt{6}}{48}K_4(3y) - \frac{\sqrt{6}}{192}K_4(6y). \tag{15}$$

The derivation above is the proof route; independently, (15) holds [VERIFIED: 40 digits, `verify/beta4_lambert.py`, error $< 10^{-35}$ at `mp.dps`= 40]. **[TODO: [6] labels the $\beta(4)$ package as "formal or exact consequences of the displayed formulas", while the degree-four elimination and the full rank-16 cyclicity are exact-computational certificates whose modular-function/divisor upgrade remains to be written. (14) and (15) are stated here at the strength of the displayed Eichler computation; the eta-transformation input $F(W_{24}\tau) = -12i\tau F(6\tau)$, $F = \eta(2\tau)^{10}/(\eta(\tau)^4\eta(4\tau)^4)$, is proved, but the exchange constants $\frac{i\sqrt{6}}{16}$, $\frac{8i\sqrt{6}}{3}$ are cited from the dossier calculation and should be re-derived in the paper.]**

2.6. **Free integrations and the denominator exponent.** The nominal cost of (5) is d_n^r. It can be less. For each Cooper cubic s_7, s_{10}, s_{18} the project proves $(n + 1) \mid A_n$, so the first nominal Eichler primitive is arithmetically free and $d_n^2 B_n \in \mathbb{Z}$ rather than $d_n^3 B_n \in \mathbb{Z}$: in the scalar case exactly a one-unit reduction of the CDT denominator penalty, $\tau_F : 3 \mapsto 2$. The weight-three cuspidal row (9) has $k = 2$ for the structural reason that its source has weight 3, not 4. Since k enters the design rule of §5.3 linearly and $\log \Lambda$ only logarithmically, a unit of k is worth more than any plausible gain in Λ; this is why §2.6 is not a footnote. A general exactness criterion for Cooper-type free integrations is [OPEN].

2.7. **Evidence summary for this section.**

Claim	Class
$B_n = [t^n]F\theta_q^{-r}\Phi$, $\Phi = t\sigma^r/(P_rF)$, all fifteen, all n (Thms. 2.4, 2.5)	[PROVED] [11]
$\Phi = F\theta_q t$ (Prop. 2.6)	[PROVED]
Table 1; zero cuspidal projection (Thm. A)	[PROVED] [2]
Same identifications, independent finite check to q^{60} with the fit frozen at q^{26}	[VERIFIED: q^{60}, held-out q^{27}–q^{60}]
No weight-3/weight-4 fit for s_7, s_{10}, s_{18} at levels $\{7, 10, 18\}$ and divisors	[EXCLUDED: stated bases]
Fold connection lemma (Lem. 2.10)	[PROVED]
Nine known limits recovered from the identified source	[VERIFIED: numeric, $N = 26$, tracked]
Uniform closed evaluation of (8) giving the rational prefactor	[OPEN]
$\mathbf{B}, \delta, \eta$ are exactly the complex-root families	[PROVED]
Complex Eichler values to 15/39/33 digits; no relation over the stated bases	[VERIFIED: $N = 60$, dps 80] / [EXCLUDED: basis]
Domb apparatus: $f^*(\tau_{12}) = 0$, $f^*\sqrt{1-4t} = \Phi_\alpha$, $d_n^3 B_n^* \in \mathbb{Z}$, $\xi^* = \frac{1}{2}L(f_6, 3)$ (Thms. 2.13, 2.14)	[PROVED] [8]
Weight-three row (9): recurrence, $k = 2$, limit $L(f, 2)$	[VERIFIED: $n \le 208$; $n \le 200$; 60 digits]
Theorems 2.15, 2.18, Props. 2.16, 2.19	[PROVED]
$\beta(4)$ CM value (14) and Lambert identity (15)	[PROVED] route; [VERIFIED: 40 digits]
$(n+1) \mid A_n$ for Cooper's three, hence $d_n^2 B_n \in \mathbb{Z}$	[PROVED]

Nothing above is claimed beyond its class. In particular the twelve-family source table is a theorem [2]; the q^{60} computation [3] is an independent check of the same statement and is reported as such, not as its justification.

3. Archimedean geometry and the single-row score

3.1. **Singular points and rates.** Let L be the Picard–Fuchs operator of f in the coordinate t (Section 2). Its singular points in the finite t-plane are the images of the cusps and elliptic points of Γ together with the zeros and poles of any rational factor in f; the latter are apparent. Order the non-apparent singularities by modulus, $0 < |t_1| \le |t_2| \le \cdots$. Then $a_n \asymp |t_1|^{-n}$ up to polynomial factors, so $\lambda_1 = 1/|t_1|$. The companion b solves the inhomogeneous equation and $b - \xi a$, for the Apéry limit ξ, is the unique combination continuing analytically past t_1; its growth is governed by the next singularity, $\lambda_2 = 1/|t_2|$. Hence

$$\text{score} := \log(1/|\lambda_2|) - k, \qquad \xi \notin \mathbb{Q} \ \text{ if } \ \text{score} > 0,$$

where k is the denominator exponent, $d_n^k b_n \in \mathbb{Z}$.

When a Fricke involution acts on t by $t \mapsto 1/(ct)$ the singular points pair off, $t_1 t_2 = 1/c$, and the recurrence is palindromic: $\lambda_1 \lambda_2 = c$. For Apéry's two rows $c = \pm 1$ and $\lambda_2 = 1/\lambda_1$; this is the precise sense in which levels 5 and 6 are perfect. For Zagier's row E (level 8, Catalan) the singularities are $t = \frac{1}{8}, \frac{1}{4}$, not reciprocal, and $\lambda_2 = 4 > 1$: the ratio b_n/a_n converges (2^{-n}) but the linear form diverges. This is the geometric reason every scalar modular Catalan system in the literature fails, and the reason the Catalan work of Section 5 uses two rows.

3.2. **Denominators.** Each Eichler integration D^{-1} costs at most one factor d_n [8], so generically $k = w + 1$. The exception is a *free integration*: for Cooper's s_7, s_{10}, s_{18} one has $(n + 1) \mid A_n$ [16], hence $d_n^2 B_n \in \mathbb{Z}$ instead of d_n^3. Sharpness of k was verified for every row in Table 2 to $n = 200$.

3.3. **The census.** Table 2 records, for every row considered in this paper, the period, c, $\lambda_{1,2}$, k, the slopes and denominator rates of Section 4, the score, and the *budget* $\log \lambda_1 - k + \sum_p \kappa_p \log p$ (the score a row would have if all its p-adic resource could be harvested; Section 5). Two rows have positive score: Apéry's. Three have no archimedean limit at all (complex-conjugate roots: Zagier B, Almkvist–Zudilin $(7, 3, 81)$ and $(11, 5, 125)$), yet each has a positive p-adic slope and, in B's case, a p-adic limit shared with C.

Conjecture 3.1 (Scalar barrier)**.** *For trivial character and $w \geq 3$, every integral Hauptmodul system of* Sym^w *type on a genus-zero group commensurable with $SL_2(\mathbb{Z})$ has negative score.*

The evidence is the level-≤ 25 eta-quotient census of [9], in which level 6 is the unique winner for $w = 2$, and the absence of any positive-score $\zeta(5)$ or $\zeta(7)$ system among the many constructed in the project (e.g. the level-12 $\zeta(5)$ system, whose conditional function has a $\log^5$ singularity *on the unit circle*, so that its linear form decays only like $(\log n)^4/n$). Raising the level crowds cusps into the unit q-disc and shrinks $|t_2|$ while the required $k = w + 1$ grows.

4. p-adic slopes and extension-class rigidity

Throughout this section a *row* is a pair of sequences (a_n, b_n) solving one three-term recurrence

$$(n + 1)^{w+1} u_{n+1} = P(n)\, u_n - c\, n^{w+1} u_{n-1}, \qquad a_0 = 1, \quad b_0 = 0,\ b_1 = 1, \tag{16}$$

with $P \in \mathbb{Z}[n]$, $c \in \mathbb{Q}^\times$, $w \in \{1, 2\}$; for $w = 1$ this is Zagier's normalisation $P(n) = an^2 + an + b$, for $w = 2$ the Almkvist–Zudilin normalisation $P(n) = (2n + 1)(an^2 + an + b)$. The constant c is the product of the two characteristic roots, $c = \lambda_1 \lambda_2$.

4.1. The slope law.

Proposition 4.1 (Casoratian)**.** *For every row of the form* (16)*,*

$$a_n b_{n-1} - a_{n-1} b_n = -\frac{c^{n-1}}{n^{w+1}} \qquad (n \geq 1).$$

Proof. Write $W_n = a_n b_{n-1} - a_{n-1} b_n$. Multiplying (16) for a by b_n and for b by a_n and subtracting gives $(n + 1)^{w+1} W_{n+1} = c\, n^{w+1} W_n$, and $W_1 = a_1 b_0 - a_0 b_1 = -1$. □

Definition 4.2. Let p be prime. Say the row has *denominator rate* $\kappa_p \geq 0$ at p if $v_p(a_n) = -\kappa_p n + O(\log n)$, and *$p$-adic slope* σ_p if

$$v_p\Big(\frac{b_n}{a_n} - \frac{b_{n-1}}{a_{n-1}}\Big) = \sigma_p n + O(\log n).$$

Proposition C (C: slope law). [PROVED] If the row has denominator rate κ_p at p then it has p-adic slope

$$\sigma_p = v_p(c) + 2\kappa_p.$$

In particular, if $\sigma_p > 0$ the ratios b_n/a_n converge in $\mathbb{Q}_p$ to a limit ξ_p, with $v_p(\xi_p - b_n/a_n) = \sigma_p n + O(\log n)$.

TABLE 2. Census of lattice rows: parameters (a, b, c) (or defining recurrence), order $w + 1$, level/group where known, archimedean period, $c = \lambda_1\lambda_2$, characteristic roots $\lambda_{1,2}$ (4 s.f.; for complex-conjugate pairs, $|\lambda_1| = |\lambda_2| = \sqrt{|c|}$ is listed once), denominator exponent k (sharp where marked ✓, verified $n \le 200$), measured p-adic slopes σ_p at $n \approx 300$ ("–" = no growing slope; predicted value $v_p(c) + 2\kappa_p$ in parentheses where it differs from the naive $v_p(c)$), denominator growth rate κ_p of a_n where $a_n \notin \mathbb{Z}$, single-row score $\log(1/|\lambda_2|) - k$, and budget $\log \lambda_1 - k + \sum_p \kappa_p \log p$. Companion normalisation $b_0 = 0, b_1 = 1$ throughout; see README for the few rows using a different convention.

name	(a,b,c)/def.	w+1	level	period	c	λ_1	λ_2	k	σ_2
Zagier A	$(7,2,-8)$	2	4/6	$\zeta(2)/4$	-8	8.000	-1.000	2✓	3(3)
Zagier B	$(9,3,27)$	2	9	none (complex)	27	5.196±		2✓	0
Zagier C	$(10,3,9)$	2	6/9	$L(2,\chi_{-3})/2$	9	9.000	1.000	2✓	0
Zagier D	$(11,3,-1)$	2	11	$\zeta(2)/5$	-1	11.09	-0.0902	2✓	0
Zagier E	$(12,4,32)$	2	8	$G/2$ (Catalan)	32	8.000	4.000	2✓	5(5)
Zagier F	$(17,6,72)$	2	6/8	$5L(2,\chi_{-3})/8$	72	9.000	8.000	2✓	3(3)
AZ$(7,3,81)$	$(7,3,81)$	3	Beauville-IV	none (complex)	81	9.000±		3✓	0
AZ$(9,3,-27)$	$(9,3,-27)$	3	–	$L(3,\chi_{-3})/3$ †	-27	19.39	-1.392	3✓	0
Domb $(10,4,64)$	$(10,4,64)$	3	6	$7\zeta(3)/24$	64	16.00	4.000	3✓	6(6)
$\eta = (11,5,125)$	$(11,5,125)$	3	order-6	$\frac{1}{2}L(3,\chi_5)$ †	125	11.18±		3✓	0
$T = (12,4,16)$	$(12,4,16)$	3	6	$7\zeta(3)/32$	16	23.31	0.6863	3✓	4(4)
Apéry $(17,5,1)$	$(17,5,1)$	3	5/6	$\zeta(3)/6$	1	33.97	0.02944	3✓	0
Cooper s_7	A_1=4, order-3 rec.	3	7	$\zeta(2)/7$	-27	27.00	-1.000	2✓	–
Cooper s_{10}	A_1=2, order-3 rec.	3	10	$\zeta(2)/5$	-64	16.00	-4.000	2✓	–
Cooper s_{18}	A_1=6, order-3 rec.	3	18	$\frac{1}{2}L(2,\chi_{-3})$	192	16.00	12.00	2✓	–
Zudilin (Catalan)	Q_0=1, Q_1=$\frac{7}{4}$, ord.-2	2	–	G	1	11.09	0.0902	n/a‡	8(8)
Nesterenko $(4,7)$	Padé det. construction	–	–	G	–§	–§	–§	–§	–§
$L(f,2)$, wt. 3 cusp	Domb curve, a_1=2	2	12,χ_{-3}	$L(f,2)$	64	16.00	4.000	2✓	2¶(6)
$\zeta(7)$, level 24	Sym^6, order-7	7	24	$\frac{1463}{13824}\zeta(7)$	–‖	6.828	4.828	?‖	none

†Period identification cited from `consolidation/SLOPE_CENSUS.md` §2 (archive source-text identification for AZ$(9,3,-27)$; book `v8/08c_chi5_application.tex` for η), not independently re-derived by `lindep` in this pass.
‡The Zudilin row is non-integral ($a_n = Q_m \notin \mathbb{Z}$); the fixed-$k$ column does not apply and $\kappa_2 = 4$ carries the denominator cost instead (verified: $v_2(Q_m) = -4m + 2s_2(m)$ to $m \le 40$). Its own-row score is likewise not of the fixed-k form; the budget $+1.18$ is cited from `consolidation/THEORY_NOTES_03_lattices.md` §5 rather than recomputed with a newly-fixed convention here (flagged in `README.md`). §Nesterenko $(4,7)$ and the Zudilin budget: $\lambda_{1,2}, k, \sigma_p$ are cited from `consolidation/CATALAN_AUDIT.md` (exact rows verified there: $\log B_n/n \to 7.6507$, $\log|4B_nG - C_n|/n \to -7.3109$, $\log|V_nG - U_n|/n \to E_2$=14.3931) and were *not* independently decomposed into a single $(\lambda_1, \lambda_2, k)$ triple within this task; see `README.md` for the honest limitation. ¶Measured $\sigma_2 = 2$ for $L(f,2)$ disagrees with the naive prediction $v_2(64) = 6$ quoted in `consolidation/SPORADIC_SEARCH.md`; a_n is confirmed integral ($\kappa_2 = 0$) to $n = 300$, so the discrepancy is not a denominator effect. The budget $\log\lambda_1 - k = 0.7726$ (independent of σ_2) matches `SPORADIC_SEARCH.md` exactly. See README. ‖The order-7 recurrence for the level-24 $\zeta(7)$ row was not fit in closed form (coefficients not fully available in the source notes); k, c, score and budget are therefore left undetermined. $\lambda_{1,2}$ are the paper's own stated values ($4 + 2\sqrt{2}$ and its $1/\sqrt{2}$-scaled partner), confirmed empirically via the decay rate of B_n/A_n − target to 33 digits at $n = 218$.

Proof. By Proposition 4.1, $v_p(b_n/a_n - b_{n-1}/a_{n-1}) = (n-1)v_p(c) - (w+1)v_p(n) - v_p(a_n) - v_p(a_{n-1})$, and the last two terms contribute $2\kappa_p n + O(\log n)$. Ultrametric convergence follows when $\sigma_p > 0$; the tail valuation is the minimum of the remaining increments. □

Corollary 4.3 (Product formula). *For an integral row ($\kappa_p = 0$ for all p),*

$$\sum_p \sigma_p \log p = \log|\lambda_1\lambda_2|.$$

Thus the archimedean defect $\log|\lambda_1\lambda_2|$ of a row—the amount by which it fails to be "Apéry-perfect" ($\lambda_1\lambda_2 = \pm 1$, as for both of Apéry's rows)—is exactly redistributed as p-adic convergence at the primes dividing c. For modular rows c is a product of values of the Hauptmodul at cusps and elliptic points, so $\sigma_p > 0$ occurs only at primes of the level. Non-integral (hypergeometric) rows have an additional resource κ_p which the product formula does not see: Zudilin's Catalan row has $c = 1$, $\kappa_2 = 4$ and slope $\sigma_2 = 8$ [31]. Table 2 lists c, σ_p and κ_p for every row considered in this paper; in every case the measured slope agrees with Proposition C to within $O(\log n)/n$.

Remark 4.4 (Non-Zagier normalisations). If the trailing coefficient of the recurrence is $c\,Q(n)$ with Q monic of degree $w+1$ but $Q(n) \neq n^{w+1}$, the Casoratian is $-c^{n-1}\prod_{k<n} Q(k)/(n!)^{w+1}$ and the slope acquires the extra term $\lim \frac{1}{n} v_p\big(\prod_{k\le n} Q(k)/(n!)^{w+1}\big)$, which can be negative. For the weight-three cusp row of Section 2, $Q(n) = (n-\frac{1}{2})^2$, the Casoratian is $\propto 4^n\binom{2n}{n}^2$ and $\sigma_2 = 2$, although $c = 64$. The product formula then reads $\sum_p \sigma_p \log p = \log|\lambda_1\lambda_2| - \lim \frac{1}{n}\log\big|\prod Q(k)/(n!)^{w+1}\big|_\infty^{-1}$ and is no longer a statement about $\lambda_1\lambda_2$ alone.

4.2. **Rigidity of p-adic limits.** Let $\Lambda \in \mathbb{R}$ be a period and let (a_n, b_n), (a'_n, b'_n) be two rows with $b_n/a_n \to r\Lambda$ and $b'_n/a'_n \to r'\Lambda$, $r, r' \in \mathbb{Q}^\times$. Define the *cross determinant*

$$\Delta_n = r\,a_n b'_n - r'\,a'_n b_n.$$

If both rows have positive slope at p with limits ξ_p, ξ'_p, then $v_p(\Delta_n)$ is bounded if $r\xi'_p \neq r'\xi_p$, and $v_p(\Delta_n) = \min(\sigma_p, \sigma'_p)\,n + O(\log n)$ if $r\xi'_p = r'\xi_p$.

Conjecture D (D: rigidity). If two rows have archimedean limits $r\Lambda$, $r'\Lambda$ with the same period Λ, and both have positive slope at p, then their p-adic limits satisfy $r\xi'_p = r'\xi_p$; equivalently there is a single $\Lambda_p \in \mathbb{Q}_p$ with $\xi_p = r\Lambda_p$, $\xi'_p = r'\Lambda_p$.

Computer-assisted finite verification 4.5. Conjecture D holds, in the sense that $v_p(\Delta_n) \ge \min(\sigma_p, \sigma'_p)n - O(\log n)$ for all $n \le N$, in the following cases (scripts in `lattice/`):

(1) $\Lambda = L(2, \chi_{-3})$, $p = 3$: Zagier's rows B, C, F ($r = \frac{1}{2}, \frac{1}{2}, \frac{5}{8}$), pairwise, $N = 300$, $v_3(\Delta_n) = 2n + O(1)$; and Cooper's s_{18} against B and C.
(2) $\Lambda = \zeta(3)$, $p = 2$: Domb $(10, 4, 64)$ ($r = \frac{7}{24}$) and $(12, 4, 16)$ ($r = \frac{7}{32}$), $N = 400$, $v_2(\Delta_n) = 4n + O(1)$.

Conversely, for rows with different periods the cross determinant has bounded valuation even when both rows have slope at p (Zagier A, E, F at $p = 2$: $v_2(\Delta_n) = 4$ for $n \le 260$), and Apéry's rows (slope 0) align with nothing.

Two features deserve emphasis. First, Zagier's row B has complex-conjugate characteristic roots, so b_n/a_n has *no* archimedean limit; yet its 3-adic limit coincides with that of row C. The p-adic limit is an invariant of the extension class that survives when the archimedean Apéry limit does not exist. Second, rigidity is a statement about *slope primes* only: the project's tower limits at unramified primes $p \ge 5$ exhibit no relations between rows sharing a period [24], so the naive "relations transfer to every realisation" is false.

The conceptual explanation is that the companion b is the Eichler integral of an Eisenstein class, so ξ_p is the p-adic realisation of the same extension class whose archimedean realisation is Λ—for the Catalan row, Calegari's overconvergent construction identifies it with a 2-adic L-value [17]. Proving Conjecture D for the Zagier rows by that method is Problem 7.1.

4.3. **Slope-free systems.** Every system obtained by full period annihilation in Section 2 is Fricke-symmetric and has $c = \pm 1$; such systems have slope 0 at every prime. This was verified directly for the level-24 $\zeta(7)$ system of Section 2: $A_n \in \mathbb{Z}$, $d_n^7 B_n \in \mathbb{Z}$, $B_n/A_n \to \frac{1463}{13824}\zeta(7)$ (all to $n = 218$), and no p-adic convergence at $p = 2, 3, 5, 7$. The rows with p-adic slope are precisely the archimedeanly *imperfect* ones. This complementarity is the starting point of Section 5.

5. Two-row constructions: the master formula and the design rule

A single Apéry row proves irrationality when its linear form decays faster than its denominators grow. When no single row does, one may try to combine two rows carrying the *same* period. This section states the closed-form quality of such a construction (Theorem E), derives from it a design rule that decides in three numbers whether a candidate pair can work, and applies it to the four cases where the arithmetic has been computed. Nothing here is an irrationality theorem; the calibration values are all < 1, and §5.5 explains structurally why.

5.1. **The correlated congruence lattice.** We recall the construction of the Catalan papers [10, 7] in the form used below. Let $\alpha \in \mathbb{R}$ be the common period. Two integer rows are given,

$$(X_{1,n}, Y_{1,n}),\ (X_{2,n}, Y_{2,n}) \in \mathbb{Z}^2, \qquad \lambda_{r,n} = X_{r,n}\alpha - Y_{r,n},$$

obtained from the two Apéry rows after sampling (row 1 at index αn, row 2 at γn), clearing denominators by a common factor S_n with $\log S_n = Kn + o(n)$, and applying any extra integralizing multiplier. Write A_r, E_r for the exponential rates of $X_{r,n}$ and $\lambda_{r,n}$. The mixed minor, after removing the common S_n from the first coordinates ($\alpha_{r,n} = X_{r,n}/S_n$),

$$h_n = \alpha_{1,n}Y_{2,n} - \alpha_{2,n}Y_{1,n},$$

is the *cross determinant*, and the arithmetic input of the method is a *proved* divisor $T_n \mid h_n$ with $\frac{1}{n}\log T_n \to G$. Setting $M_n = S_nT_n$, the *correlated congruence lattice* is

$$K_n = \Big\{(c_1, c_2) \in \mathbb{Z}^2 :\ \alpha_{1,n}c_1 + \alpha_{2,n}c_2 \equiv 0\ (T_n), \quad Y_{1,n}c_1 + Y_{2,n}c_2 \equiv 0\ (M_n)\Big\}, \tag{17}$$

so that $q(c) = (c_1X_{1,n} + c_2X_{2,n})/M_n$ and $p(c) = (c_1Y_{1,n} + c_2Y_{2,n})/M_n$ are integers. A two-line Smith-normal-form computation gives

$$[\mathbb{Z}^2 : K_n] = \frac{S_nT_n^2}{\gcd(h_n, T_nY_{1,n}, T_nY_{2,n}, S_nT_n\alpha_{1,n}, S_nT_n\alpha_{2,n}, S_nT_n^2)} \ \le\ S_nT_n = M_n$$

because $T_n \mid h_n$; passing to a sublattice $L_n \subseteq K_n$ with $M_n \le \operatorname{covol}(L_n) < 2M_n$, the division modulus and the coefficient-lattice covolume have the same exponential rate

$$\sigma = K + G. \tag{18}$$

This is the whole point of the word "correlated": the determinant divisibility is used in two congruences at the cost of one copy of T_n rather than T_n^2.

The selection step is the two-successive-minima theorem of [7, Thm. 5.1], stated there for an arbitrary real α and used here as a black box.

Theorem 5.1 (two-minimum selection; [7, Thm. 5.1])**.** [PROVED] *Suppose $S_n \mid X_{1,n}, X_{2,n}$, the lattice and output rows are as above with $d_n = [\mathbb{Z}^2 : L_n]$, the six two-sided rate statements hold, $s_n = \frac{1}{n}\log S_n \to K$ with $0 \le \kappa_n = \frac{1}{n}\log d_n \le s_n$, the* crossed gap $A_2 + E_1 > A_1 + E_2$ *holds, and H_n, F_n defined by*

$$x_n = \frac{\kappa_n + E_2 - E_1}{2}, \qquad H_n = \kappa_n - x_n + A_2 - s_n, \qquad F_n = x_n + E_1 - s_n$$

are eventually bounded away from zero. Then for all large n one can select $c_n \in L_n$ with $q_n = q(c_n) \neq 0$, $\ell_n = q_n\alpha - p_n \neq 0$,

$$\log|q_n| \ge H_nn - o(n), \qquad \frac{\log|\ell_n|}{\log|q_n|} \le \frac{F_n}{H_n} + o(1),$$

hence worthiness $\delta \ge 1 - F_n/H_n$.

The proof applies Minkowski's second theorem to the anisotropic rectangle $|u| \le e^{x_n n}$, $|v| \le e^{(\kappa_n - x_n)n}$ (area $4d_n$), takes vectors attaining the two successive minima, and splits into three cases according to the size of $r_n = -\frac{1}{n}\log \mu_{1,n}$ and whether the first minimum's linear form vanishes. The use of two minima rather than Minkowski's first theorem alone is what prevents an exceptionally short *rationality-kernel vector* from spoiling the lower bound on $|q_n|$.

Remark 5.2 (the hypothesis $F > 0$ is not decorative). Hypothesis (4) of Theorem 5.1 — H_n and F_n eventually bounded below by a *positive* constant — is exactly what fails if one pushes σ past $E_1 + E_2$. In that regime $F_n < 0$ and the method degenerates: the box area equals the covolume, so Minkowski's first theorem in that box *is* Dirichlet's theorem, which produces such pairs for every real number, rational ones included. The proof's non-vanishing $\ell_n \neq 0$ comes from the output-determinant identity, which says only that the two minima's forms are not *both* zero; when the first minimum has $\ell = 0$ — precisely the rationality-kernel case — the argument switches to the second vector, which is nonzero but carries no $e^{F_n n}$ upper bound. The construction can deliver "$\ell_n \neq 0$" or "$|\ell_n| \le e^{F_n n}$", never both. This was verified by an explicit control experiment: replacing the period G by a rational $G^* = \mathrm{bestappr}(G, 10^{320})$ leaves the lattice (17) unchanged (it never uses G) and reproduces the same numerics, $|q_n G - p_n| = e^{-78.051}$ against $|q_n G^* - p_n| = e^{-78.050}$ at $n = 98$, with no contradiction because $1/\operatorname{den}(G^*) = e^{-735.7}$ [22, §4]. Everything below is therefore stated in the regime $F > 0$, where the conclusion is a worthiness statement and nothing more.

5.2. **Theorem E: the master formula.** Doing the bookkeeping of §5.1 in closed form gives one formula that covers every instance computed in this project.

Theorem E (master formula)*.* [PROVED] modulo Theorem 5.1. Consider a two-row construction in which

- the *decayer* is sampled at γn, with $|a_n| \sim \Lambda_{\mathrm{dec}}^{\gamma n}$ and linear form $\sim \lambda^{\gamma n}$, $\lambda < 1$, and p-adic slope σ_{dec};
- the *engine* is sampled at αn, with linear form $\sim (\rho_2^{\mathrm{eng}})^{\alpha n}$ (which may grow) and slope σ_{eng};
- the common denominator clearing has rate $S = k \cdot \max(\alpha, \gamma)$, and any extra integralizing multiplier has rate η per unit of n;
- the hidden divisor rate is $G = \sum_p \min(\sigma_p^{\mathrm{eng}}\alpha,\ \sigma_p^{\mathrm{dec}}\gamma)\log p$.

Then, in the notation of Theorem 5.1,

$$\boxed{F = \tfrac{1}{2}\Big[\underbrace{S + \eta + \alpha\log\rho_2^{\mathrm{eng}}}_{\text{COST}} - \underbrace{\big(\gamma\log\tfrac{1}{\lambda} + G\big)}_{\text{RESOURCE}}\Big], \qquad H = F + \gamma\log\frac{\Lambda_{\mathrm{dec}}}{\lambda}, \qquad \delta = 1 - \frac{F}{H}} \tag{19}$$

and $H - F = A_{\mathrm{dec}} - E_{\mathrm{dec}}$. In words: *$F$ is exactly one half of the cost–resource deficit.*

Proof. Immediate from $x = \frac{1}{2}(\sigma + E_2 - E_1)$, $H = A_2 - x$, $F = x + E_1 - \sigma$ of Theorem 5.1 with $\sigma = K + G$ as in (18), after substituting the four rates $A_{\mathrm{eng}} = S + \eta + \alpha\log\Lambda_{\mathrm{eng}}$, $E_{\mathrm{eng}} = S + \eta + \alpha\log\rho_2^{\mathrm{eng}}$, $A_{\mathrm{dec}} = S + \gamma\log\Lambda_{\mathrm{dec}}$, $E_{\mathrm{dec}} = S + \gamma\log\lambda$, and simplifying. (The two-row roles are labelled so that the decayer plays the part of row 2 in Theorem 5.1.) The multi-prime form of G is legitimate because the correlated-lattice step of §5.1 needs only $T_n \mid h_n$ and nothing about T_n being a prime power; the Smith-normal-form index bound goes through verbatim for $T_n = \prod_p p^{g_p n}$. $\square$

Remark 5.3 (calibration). Formula (19) reproduces every independently derived quality number in the project.

instance	F	H	δ	independent s
Catalan, Zudilin $\times$ E at 3:6	$6 - \frac{15}{2}\log\varphi$	$6 + \frac{45}{2}\log\varphi$	0.8579144525	[10] lineage
Catalan, Zudilin $\times$ E at 5:8	2.5985578256	26.6591490786	0.9025266029	[10]
$\zeta(3)$, Domb $\times$ T at 2:3	1.1627320584	11.7392151026	0.9009531687	[25]
Catalan, Zudilin $\times$ Nesterenko, no bridge	7.746367029	22.707995624	0.6588705072	[4]
same pair, at the Lean threshold $\sigma = 12 + k^* \log 2$	0	14.961629	1	[13]

The last line is the knife-edge: $F = 0$ exactly at $k = k^*$, so the Lean development formalizes precisely a *family* of constructions,

$$\delta(k) = \frac{D}{D + F(k)}, \qquad D = A_N - E_N, \qquad F(k) = \frac{\log 2}{2}(k^* - k), \qquad 0 \le k < k^*,$$

with $\delta(k) \to 1$ as $k \uparrow k^*$ and $\delta(22) > 0.99$. The rates there are $A_Z = 12 + 12\log 2 + 15\log\varphi$, $E_Z = 12 + 12\log 2 - 15\log\varphi$ for the Zudilin row at $3n$ over $S_n = D_{6n}^2$, and $A_N = 12 + 14\log 2 + 2\log T^+$, $E_N = 12 + 14\log 2 + 2\log T^-$ for the Nesterenko $(4,7)$ row, with $T^\pm = (3303 \pm 437\sqrt{57})/144$ and $T^+T^- = 32/27$; numerically $A_N = 29.354774$, $E_N = 14.393145$, $1 - E_N/A_N = 0.5096830$, and

$$k^* = \frac{E_Z + E_N - 12}{\log 2} = 22.3512905953\ldots\ , \qquad 22 < k^* < 24.$$

The formalization introduces no axiom: the Nesterenko archimedean rates and the 2-adic data (in particular the exact tail valuation $v_2(4B_n\mathcal{G}_2 - C_n) = 14n + 2 - s_2(n) - s_2(3n)$, i.e. the identification of the Nesterenko row's 2-adic limit with the Zudilin row's) are first carried as a structure `NesterenkoFormInputs` and then instantiated, in `NesterenkoUnconditional.lean`, by theorems proved in the project; the main theorem `catalan_worthiness_one_sub_eps_nesterenko` depends only on `propext`, `Classical.choice`, `Quot.sound`. *No claim of irrationality follows*: the constructions are a family, not a single sequence of worthiness 1.

5.3. **The design rule.** Formula (19) has three inputs one can shop for. Specializing it turns the search for a two-row pair into a one-line test.

Suppose first that both rows are *integral*, so $\eta = 0$ and $\kappa_p = 0$ below, and take the balance point $\alpha\sigma_{\mathrm{eng}} = \gamma\sigma_{\mathrm{dec}}$ at which the two p-adic tails have equal slope (so that the minimum in G is attained twice). Then for a single slope prime p with $\log|c| = \sigma_p \log p$, the product formula $\Lambda\lambda = c$ of §4 gives $\gamma\log\frac{1}{\lambda} + G = \gamma\log\Lambda_{\mathrm{dec}}$ *exactly*, and (19) collapses to

Proposition 5.4 (design rule, integral rows)**.** [PROVED] *With the above hypotheses,*

$$\boxed{\delta > 1 \iff \log\Lambda_{\mathrm{dec}} > k\max(r,1) + r\log\rho_2^{\mathrm{eng}}, \qquad r = \frac{\alpha}{\gamma} = \frac{\sigma_{\mathrm{dec}}}{\sigma_{\mathrm{eng}}}.} \tag{20}$$

What the decaying row contributes is therefore its *numerator growth* $\log\Lambda$, not its archimedean decay: the p-adic slope makes up the whole difference, automatically. The method is a machine for converting the product formula into Diophantine decay. The quantity $\log\Lambda - k$ is the *budget* of a row.

Two corrections must be attached to (20), and both matter.

(i) Order > 2. The step $\gamma\log\frac{1}{\lambda} + G = \gamma\log\Lambda$ uses $\Lambda\lambda = c$, an order-two identity. For a scalar recurrence of order m with roots $\lambda_1, \ldots, \lambda_m$ the product formula reads $\sum_p \sigma_p \log p = \log|\prod_i \lambda_i|$, so the correct row score, valid at any order, is

$$\text{score} = \log\frac{1}{|\lambda_2|} - k, \qquad \text{harvestable budget} = \text{score} + \log\Big|\prod_i \lambda_i\Big|, \tag{21}$$

which reduces to $\log|\lambda_1| - k$ only when $m = 2$. Using $\log\lambda_1 - k$ on a higher-order row overstates the budget, sometimes wildly: the level-12 $\zeta(5)$ row has harvestable budget -5, not $+5.54$. All the rows used below are of order two, so §§5.1–5.4 are unaffected.

(ii) Non-integral rows and κ. Proposition 5.4 assumed $v_p(a_n) \geq 0$. If instead the row carries p-power denominators at rate κ_p, i.e. $v_p(a_n) = -\kappa_p n + O(\log n)$, then the two $-v_p(a)$ terms in the Casoratian slope law each contribute $+\kappa_p n$, and by Proposition C the slope is

$$\sigma_p = v_p(c) + 2\kappa_p. \tag{22}$$

Clearing the denominators costs $\eta = \kappa_p \log p$ per index, so the *net* resource is $(\sigma_p - \kappa_p)\log p = (v_p(c) + \kappa_p)\log p$ and the corrected budget is

$$\boxed{\text{budget} = \log\Lambda - k + \sum_p \kappa_p \log p.} \tag{23}$$

The κ term is a second, independent source of p-adic resource, *invisible to the product formula.* An integral row is capped at $\log\Lambda - k$ as (20) says; a non-integral row is not capped there at all. The measurement that forces this: Zudilin's Catalan row has $c = \lambda_1\lambda_2 = 1$, so $v_2(c) = 0$, yet its measured 2-adic convergence slope is 8; and indeed $v_2(Q_m) = -4m + 2s_2(m)$ exactly [VERIFIED: $1 \leq m \leq 160$, zero exceptions], so $\kappa_2 = 4$ and $\sigma_2 = 0 + 2\cdot 4 = 8$. For comparison (22) gives Domb $6 + 0 = 6$, T $4 + 0 = 4$, Catalan's $\mathbf{E}$ row $5 + 0 = 5$, all matching. With $\log\Lambda = 5\log\varphi$, $\kappa_2 \log 2 = 4\log 2$ and $k = 4$, Zudilin's row has budget $+1.17864$, by far the largest of any row known — and every unit of its lead comes from κ_2.

Remark 5.5 (multi-prime bridging). The bridge generalizes to several primes for free, with $G = \sum_p \min(\sigma_p^{\text{eng}}\alpha, \sigma_p^{\text{dec}}\gamma)\log p$ (proof of Theorem E). But it does not raise the ceiling for integral rows: since $\sum_p \sigma_p^{\text{dec}} \log p = \log|c^{\text{dec}}|$ and $\log\frac{1}{\lambda} + \log|c| = \log\Lambda$, the resource is still at most $\gamma\log\Lambda_{\text{dec}}$; and the balance ratio $\alpha : \gamma$ is a single number, so at most one prime can be balanced, the others contributing their unbalanced minimum. Moreover it is *unrealizable in the census*: across the twelve modular sporadic families the values of c are $-8, 27, 9, -1, 32, 72, 81, -27, 64, 125, 16, 1$ for $\mathbf{A}, \ldots, \mathbf{F}, \delta, \zeta, \alpha, \eta, \varepsilon, \gamma$, so $\mathbf{F}$ ($c = 72 = 2^3 \cdot 3^2$) is the unique row with two slope primes. Its period $\frac{5}{8}L(2, \chi_{-3})$ is shared with $\mathbf{B}, \mathbf{C}, s_{18}$, all of which have $\sigma_2 = 0$; since the per-prime term is a minimum over the two rows, the $p = 2$ contribution vanishes for every pair containing $\mathbf{F}$. Multi-prime bridging is thus available in principle and realizable nowhere in the canonical classification — a clean negative, not an oversight.

(a) Catalan, two versions. The rows are Zudilin's Apéry-like Catalan recurrence [32, 30] — $Q_0 = 1$, $Q_1 = \frac{7}{4}$, $P_0 = 0$, $P_1 = \frac{13}{8}$, with $\frac{1}{m}\log|Q_m| \to 5\log\varphi$, $\frac{1}{m}\log|Q_m G - P_m| \to -5\log\varphi$, $2^{e_m}Q_m \in \mathbb{Z}$, $2^{e_m}D_{2m-1}^2 P_m \in \mathbb{Z}$, $e_m = 4m + O(\log m)$ — and the modular $\mathbf{E}$ row of §2, with $t = \eta_1^4\eta_4^2\eta_8^4/\eta_2^{10}$, $F = \eta_2^{10}/(\eta_1^4\eta_4^4)$, source $\Phi_{\mathbf{E}} = S - 8S(2\tau)$ where $S = E_{3,\chi_{-4},1}$, so $L(\Phi_{\mathbf{E}}, s) = (1 - 8\cdot 2^{-s})\beta(s)\zeta(s-2)$ and $L(\Phi_{\mathbf{E}}, 2) = G/2$; characteristic roots 8 and 4, $A_n \in \mathbb{Z}$, $D_n^2 B_n \in \mathbb{Z}$. The arithmetic bridge is the statement that the two rows converge to the *same* 2-adic Catalan period:

Lemma 5.6 (common 2-adic period; [10, Lem. 1.1]). *There is $G_2 \in \mathbb{Q}_2$ with*

$$v_2\Big(G_2 - \frac{P_m}{Q_m}\Big) = 8m - 1 - 4s_2(m), \qquad v_2\Big(G_2 - \frac{2B_n}{A_n}\Big) = 5n - 1 - 4s_2(n),$$

whence $v_2(P_m A_{2m} - 2B_{2m}Q_m) = 4m - 1$, and for $\Delta_{i,k} = P_iA_k - 2Q_iB_k$,

$$v_2(\Delta_{i,k}) \geq \min\big\{4i - 1 - 2s_2(i) + 2s_2(k),\ 5k - 4i - 1 + 2s_2(i) - 2s_2(k)\big\}. \tag{24}$$

The proof combines the two exact Wronskians with $v_2(Q_m) = -4m+2s_2(m)$ and $v_2(A_n) = 2s_2(n)$; the identification of the two limits with $\zeta_2(2)$ uses Rivoal's moving Padé specialization of Beukers' continued fraction on the Zudilin side [27] and the level-8 form of Calegari's overconvergent 2-adic construction on the modular side [18]. **[TODO: [10] labels this step "Idea of proof", and [21] records the identification of the two 2-adic limits as the one genuine gap in the 0.9025 paper. It should be cited as [PROVED] only after the moving-point passage is written out; until then Lemma 5.6 is the hypothesis under which Theorem E is applied to this pair.]**

The two affine slopes in (24) balance when $8i = 5k$, which is the arithmetic origin of the sampling ratio $i = 5n$, $k = 8n$; and since $s_2(8n) = s_2(n)$, (24) gives $v_2(\Delta_{5n,8n}) \geq 20n - O(\log n)$. With $e_{5n} = 20n + O(\log n)$ the total divisor rate is $G = 40 \log 2$. Feeding $\gamma = 5$, $\alpha = 8$, $k = 2$, $S = 2\max(10, 8) = 20$, $\eta = \kappa_2\gamma \log 2 = 20 \log 2$, $\rho_2^{\mathrm{eng}} = 4$, $\lambda = \varphi^{-5}$, $\Lambda_{\mathrm{dec}} = \varphi^5$ into (19):

$$F = \tfrac{1}{2}\big[20+20\log 2+16\log 2-25\log\varphi-40\log 2\big] = 10-2\log 2-\tfrac{25}{2}\log\varphi, \qquad H = F+50\log\varphi = 10-2\log 2+\tfrac{75}{2}\log\varphi$$

so $\delta = (H - F)/H = 50\log\varphi/(10 - 2\log 2 + \frac{75}{2}\log\varphi) = 0.902526602856971\ldots$, reproducing [10] exactly. The same paper shows that $r = c/a = 8/5$ is the unique optimum inside this scalar architecture: with $H(a, c) = \max(2a, c) + (2a + c)\log 2 - \frac{1}{2}\min(8a, 5c)\log 2 + \frac{15}{2}a\log\varphi$ and $\delta = 10a\log\varphi/H$, $H(r)$ is strictly decreasing for $0 < r < 8/5$ and strictly increasing beyond.

At the coarser sampling $3n : 6n$ the same formula gives $\gamma = 3$, $\alpha = 6$, $S = 12$, $\eta = 12\log 2$, $\alpha\log\rho_2^{\mathrm{eng}} = 12\log 2$, $G = \min(5\cdot 6, 8\cdot 3)\log 2 = 24\log 2$, hence $F = 6 - \frac{15}{2}\log\varphi$, $H = 6 + \frac{45}{2}\log\varphi$, and

$$\delta_{ZE} = \frac{30\log\varphi}{6 + \frac{45}{2}\log\varphi} = 0.8579144524736\ldots$$

[TODO: The 0.857914 construction is reconstructed here from the master formula and from the cross-divisor statement 2^{24n} at $m = 3n$; the project's written source for it is a research conversation rather than a paper. Either the architecture ($3n{:}6n$ sampling over $S_n = D_{6n}^2$) should be written up, or the number should be presented only as a specialization of Theorem E, which is how it is presented above.]

Against the Nesterenko $(4, 7)$ row instead of **E**, with no bridge at all ($G = 0$, $\sigma = \log S = 12$), (19) gives $F = \frac{E_Z+E_N-12}{2} = 7.746367$, $H = F + (A_N - E_N) = 22.707996$, $\delta = 0.6588705$, matching the independently derived full-LCM paper value 0.6588 [4]. The empirical measurement on River's exact row CSVs to $n = 98$ gives 0.6460, drifting down; the agreement is "same construction", not a precision match, the residual being finite-n effects ($\log S_n/n = 11.86$ at $n = 98$ against its limit 12) and the use of a best-reduced-basis vector rather than the prescribed anisotropic box [25, §15]. Adding the missing second congruence to the same rows raises the measured δ from 0.646 to ≈ 1.01, converging to 1 from above — but $\log|q_nG - p_n|$ stays between -13 and -23 for all n tested and does *not* tend to $-\infty$. That is worthiness 1, the knife-edge $F = 0$; it is not irrationality, and it agrees exactly with the formalization.

(b) $\zeta(3)$: Domb × T. This is the calibration case with a known answer. In the Zagier/Almkvist–Zudilin normalization $(n+1)^3u_{n+1} = (2n+1)(an^2+an+b)u_n - cn^3u_{n-1}$, take

row	(a,b,c)	$a_n \sim$	limit	linear form	$\sigma_2 = v_2(c)$
D (Domb)	$(10,4,64)$	16^n	$\frac{7}{24}\zeta(3)$	$\sim 4^n$ (grows)	6
T	$(12,4,16)$	Λ^n, $\Lambda = 12+8\sqrt{2}$	$\frac{7}{32}\zeta(3)$	$\sim \lambda_2^n$, $\lambda_2 = 12-8\sqrt{2}$ (decays)	4

with $\Lambda\lambda_2 = 16$, $a_n^{\mathbf{T}} = \sum_k \binom{n}{k}^2\binom{2k}{n}^2$ and $a_n^{\mathbf{D}} = \sum_k \binom{n}{k}^2\binom{2k}{k}\binom{2n-2k}{n-k}$. Both are integral with $d_n^3 b_n \in \mathbb{Z}$ and $k = 3$ sharp: $d_n^2 b_n \notin \mathbb{Z}$, and the measured $k_{\mathrm{eff}}(n) = \log\mathrm{den}(b_n)/n$ is 2.94 at $n = 400$ against $\log d_n^3/n = 2.98$ [VERIFIED: $n \leq 400$]. *There is no free denominator saving here.*

The Casoratian $a_n b_{n-1} - a_{n-1} b_n = -c^{n-1}/n^3$ gives exactly $v_2(b_n/a_n - b_{n-1}/a_{n-1}) = (n-1)v_2(c) - 3v_2(n) - v_2(a_n) - v_2(a_{n-1}) =: \iota(n)$ [PROVED], so b_n/a_n is 2-adically Cauchy with $v_2(\xi_2 - b_m/a_m) = \min_{j>m} \iota(j)$. For $\mathbf{T}$ everything collapses: $v_2(a_n^{\mathbf{T}}) = 3s_2(n) - [n \text{ odd}]$ and $\iota_{\mathbf{T}}(n) = 4n - 6 - 6s_2(n-1)$ [VERIFIED: $n \le 700$ resp. $n \le 699$, no exceptions]; for $\mathbf{D}$ only the bound $v_2(a_n^{\mathbf{D}}) - 3s_2(n) \in [-5, 9]$ for $n \le 700$. Note ι is not monotone here, so the "first increment" shortcut is wrong and one must take the minimum.

For the cross determinant $\Delta_{m,n} = 3a_n^{\mathbf{T}} b_m^{\mathbf{D}} - 4a_m^{\mathbf{D}} b_n^{\mathbf{T}}$, writing $\varepsilon = 3\xi_2^{\mathbf{D}} - 4\xi_2^{\mathbf{T}} \in \mathbb{Q}_2$, one has *conditionally on* $\varepsilon = 0$ the ultrametric bound

$$v_2(\Delta_{m,n}) \;\ge\; v_2(a_n^{\mathbf{T}}) + v_2(a_m^{\mathbf{D}}) + \min\{\tau_{\mathbf{D}}(m),\, 2 + \tau_{\mathbf{T}}(n)\},$$

[VERIFIED: $1 \le m, n \le 60$: 0 violations out of 3600, with exact equality in 3443 cases, the exceptions being ties]. The affine slopes $6m$ and $4n$ balance at $m : n = 2 : 3$, and then $v_2(\Delta_{2n,3n}) = 12n - O(\log n)$ with $12n - v_2(\Delta_{2n,3n}) \in [-5, 17]$ for $n \le 110$. The alignment of the two 2-adic limits, with the predicted rational factor, is recorded as $v_2(3a_n^{\mathbf{T}} b_n^{\mathbf{D}} - 4a_n^{\mathbf{D}} b_n^{\mathbf{T}}) = 4n + O(1)$ $(= \min(6,4)n)$ checked to $n = 400$ [24]. (The exact values at $n = 50, 100, \dots, 400$ are $4n-3, 4n-3, 4n-2, 4n-3, 4n-3, 4n-2, 4n, 4n-3$; the statement used throughout is $v_2(\Delta_{n,n}) = 4n + O(1)$.)

Feeding $\gamma = 3$ ($\mathbf{T}$, decayer), $\alpha = 2$ ($\mathbf{D}$, engine), $k = 3$, $S = 3\max(2,3) = 9$, $\eta = 0$, $\rho_2^{\text{eng}} = 4$, $G = 12\log 2$ into (19):

$$F = \tfrac{1}{2}\big[9 + 4\log 2 - 3\log\Lambda\big] = 1.1627320584, \qquad H = F + 3\log\tfrac{\Lambda}{\lambda_2} = 11.7392151026,$$

$$\delta = \frac{12\log\Lambda - 24\log 2}{9 + 9\log\Lambda - 20\log 2} = 0.9009531686558563\ldots \; < 1. \tag{25}$$

So this is *not* a second proof of Apéry's theorem. Numerical optimization over $r = \alpha/\gamma \in (0.4, 100)$ gives a unique maximum at $r = 2/3$, the same ratio that balances the two 2-adic tails, exactly as 5:8 did for Catalan. Dropping the hidden divisor entirely ($T_n = 1$) gives $\delta_{\text{arch}} = 0.6652671861$ at 2:3 and $\sup_r \delta_{\text{arch}} = 0.7278$ (approached as $r \to \infty$, not attained), so the 2-adic mechanism is worth $+0.2356860$ in quality: real and large, and not large enough. Two exact thresholds quantify the shortfall. With everything else fixed, $\delta \ge 1$ iff $k \le k^* = \log(12 + 8\sqrt{2}) - \frac{4}{3}\log 2 = 2.2248452944$, whereas $k = 3$ is sharp: an arithmetic-method saving would have to remove 26% of the denominator. With $k = 3$, one needs $\frac{1}{n}\log T_n \ge 9 + 16\log 2 - 3\log\Lambda = 10.6432$ instead of $12\log 2 = 8.3178$, i.e. an extra $2^{3.3549n}$, equivalently a Domb 2-adic slope of 7.68 instead of 6. Nothing in sight supplies either. Finally, $F > 0$ here means the linear forms *grow*: there is no irrationality statement at all, not even a weak one, and no irrationality measure. For reference, even $\delta = 1.033$ would give $\mu(\zeta(3)) \le 31.4$, worse than Apéry's 13.42 and far from Rhin–Viola; the only possible value of this architecture would be as a new proof, never as a measure.

(c) $L(2, \chi_{-3})$: four aligned rows and no decayer. Here the p-adic side is rich and the archimedean side is empty. At $p = 3$ the rows $\mathbf{B}, \mathbf{C}, \mathbf{F}$ (and s_{18}) share the period $L(2, \chi_{-3})$ with slopes $3, 2, 2$, and the cross-determinant valuations grow like $\min(\sigma_p, \sigma_p')n$: $\mathbf{B}, \mathbf{C}$ give $199, 398, 600$ at $n = 100, 200, 300$; $\mathbf{C}, \mathbf{F}$ give $198, 401, 601$; $\mathbf{B}, \mathbf{F}$ give $199, 398, 600$ — all $2n$ [VERIFIED: $n \le 300$]. Rows with different periods do not align even at a prime where both have slope ($\mathbf{A}, \mathbf{E}$ at 2: values $4, 4$, no slope). But none of the four decays: $\mathbf{C}$ has $\lambda_2 = 1$ exactly (polynomial-size linear form), $\mathbf{F}$ has $\lambda_2 = 8$, $\mathbf{B}$ has complex-conjugate roots and oscillates, and the level-12 fully purified row of §2.4 (coefficients $1, -7, 15, -57, 159, \dots$) has an order-two, degree-nine Picard–Fuchs operator with two cusps on $|t| = 1/3$ and *no fold*, so it is not the decaying partner either. Consistently, CDT's arithmetic-holonomy proof was the first [19]. The negative is useful: for $L(2, \chi_{-3})$ the obstruction is the absence of a decayer, not the budget.

(d) What the census says. Ranking every row now known by the corrected budget (23):

row	κ_2	budget	same-period decayer?
Zudilin (Catalan decayer)	4	$+1.1786$	is the decayer
weight-three cusp row, $L(f,2)$, $f=\eta_2^3\eta_6^3$	0	$+0.7726$	**no — none known**
Apéry $\zeta(3)$ $(17,5,1)$	0	$+0.5255$	is its own
$\mathbf{D}$, Apéry $\zeta(2)$	0	$+0.4061$	is its own
$\mathbf{C},\mathbf{F}$ $(L(2,\chi_{-3}))$	0	$+0.1972$	**no — no row in the class has** $\boldsymbol{\lvert\lambda_2\rvert<1}$
$\mathbf{T}$ $(\frac{7}{32}\zeta(3))$	0	$+0.1490$	yes, Domb is the engine $\Rightarrow \delta=0.9010$
$\mathbf{A},\mathbf{E}$	0	$+0.0794$	$\mathbf{E}$: yes, Zudilin's row $\Rightarrow \delta=0.9025$
everything else	—	<0	—

The pattern over the complete classification is unambiguous, and it is not about budget:

Observation 5.7. The two-lattice method has produced a non-trivial construction exactly twice, and both times the decaying partner came from outside the modular world: Zudilin's hypergeometric row for Catalan, and $\mathbf{T}$ for $\zeta(3)$ (which decays only because $\lambda_2=12-8\sqrt{2}<1$ is an accident of its quadratic). Every class where the method dies — $L(2,\chi_{-3})$ with four aligned rows, the weight-three cusp row with the largest budget of all — dies for the same reason: **no decayer**, not insufficient budget.

5.5. **Why cross-world pairs.** Observation 5.7 has a structural explanation, and it is (22).

Proposition 5.8 (cross-world principle)**.** [PROVED] *from Proposition C. A modular Apéry row of the kind classified in §2 has $a_n\in\mathbb{Z}$ automatically (triangular peeling against $t\in q\mathbb{Z}[[q]]$), hence $\kappa_p=0$ at every p, hence $\sigma_p=v_p(c)$ and budget $\le\log\Lambda-k$. A hypergeometric row — Beukers/Rivoal Padé, Rhin–Viola integrals, Zudilin's and Nesterenko's constructions — acquires $\kappa_p>0$ from its factorial/binomial normalization, and is not capped there. Therefore "you need a cross-world partner" is precisely "you need a partner with $\kappa_p>0$, and only the hypergeometric world supplies it".*

Three consequences. First, the search criterion *inverts*: one should not look for rows with $p\mid c$, but for rows with large p-power *denominators*. Second, the claim that "no amount of cleverness can exceed $\gamma\log\Lambda_{\mathrm{dec}}$", which would have made the $\zeta(3)$ failure a statement about the $(12,4,16)$ motive, is true only for integral rows and is retracted here: the $\zeta(3)$ failure is a fact about the two rows we happen to have, both of which are integral — indeed $v_2(a_n^{\mathbf{T}})\ge 0$ is *positive*, which slightly reduces the slope. Third, this yields sharp targets: a hypergeometric $\zeta(3)$ row with 2-power denominators (Rhin–Viola's $\zeta(3)$ integrals being the obvious place) paired with Domb as engine; and, for the weight-three cusp row of §2.3 with its engine parameters $k=2$, $\sigma_2=2$, $\log\rho_2^{\mathrm{eng}}=\log 4$ (the recurrence $(n+1)^2a_{n+1}=(20n^2+10n+2)a_n-16(2n-1)^2a_{n-1}$ is not in Zagier normalisation: its trailing coefficient $64(n-\frac{1}{2})^2$ gives Casoratian $\propto 4^n\binom{2n}{n}^2$, hence slope 2 rather than $v_2(64)=6$; see Remark 4.4), a decaying partner with period $L(f,2)$, $f=\eta(2\tau)^3\eta(6\tau)^3$, and

$$\Lambda_{\mathrm{dec}}>14.8,\ 29.6,\ 160 \qquad \text{according as} \qquad \sigma_{\mathrm{dec}}=1,\ 2,\ 3,$$

by (20). The exhaustive eta-quotient census offers evidence against finding it inside the modular world: over all fourteen genus-zero levels and 520 005 pairs, the largest characteristic root attached to a known-period Apéry-like row is Apéry's own $\Lambda=12+8\sqrt{2}+\cdots=33.97$, and raising the level does not raise $\log\Lambda$ — levels $10,13,18$ contribute nothing, and levels $12,16$ only reproduce rows already present at $4,6,8,9$ [23]. Lowering $\log\rho_2^{\mathrm{eng}}$, by contrast, is the cheapest lever in (19) and has never been optimized: every construction so far took whatever engine happened to be aligned, rather than the one with the slowest-growing linear form.

6. Past the threshold: what a two-row construction cannot prove

The Zudilin×Nesterenko construction for Catalan's constant G is, in the terminology of Section 5, *supercritical*: its 2-adic cross divisibility holds at rate 24 while the break-even rate is $k^* = 22.3512\ldots$ [12]. The formal continuation of the quality formula past k^* gives $\delta(k) > 1$, and a numerical lattice reduction at $k = 23$ produces integer pairs (q_n, p_n) with $|q_n G - p_n| = e^{-0.8n}$. This section explains precisely why no irrationality statement follows, and what would.

6.1. **The rationality-kernel vector.** Let $\mathcal{K}_n \subset \mathbb{Z}^2$ be the congruence lattice of coefficient vectors (c_Z, c_N) for which the combination $(q, p) = (c_Z X_n + c_N V_n,\ c_Z Y_n + c_N U_n)/M_n$ of the two integer rows is integral. $\mathcal{K}_n$ is defined by the rows alone and does not involve G. Only the metric in which short vectors are sought—weighting the coordinate q and the linear form $qG-p$ differently—depends on G.

Proposition 6.1 (Control)**.** *Let $G^* = \mathrm{bestappr}(G, 10^{320})$. The lattice reduction that produces (q_n, p_n) returns the same pairs for G and G^*, and at $n = 98$*

$$|q_n G - p_n| = e^{-78.051}, \qquad |q_n G^* - p_n| = e^{-78.050}.$$

A rational number therefore produces the identical table. No finite portion of such data can be evidence of irrationality: the Diophantine content of "$|q_n G - p_n| \to 0$" lies entirely in the assertion that $q_n G - p_n \neq 0$, and for a rational $G = a/b$ the lattice contains the *kernel vector* $(c_Z, c_N) \propto (aV - bU, -(aX - bY))$, whose form vanishes identically. In the regime $F < 0$ the Minkowski box has area equal to the covolume, the selection degenerates to Dirichlet's theorem, and the selected vector may be the kernel vector. The selection theorem of [5] requires $F > 0$ for exactly this reason; the Lean formalisation carries it as the hypothesis `hF`, which is unsatisfiable at $k \geq k^*$ [12].

6.2. **Absorption.** One might hope to exclude the kernel vector by arithmetic. The following statements, each verified on the exact rows and proved in the indicated generality, show that every device expressible through valuations is absorbed.

Theorem 6.2 (Absorption)**.** *Let $G = a/b$ be a hypothetical rational value and $(q_n, p_n) = M_n(b, a)$ the kernel output. Then:*

(1) (Exact 2-adic content.) $v_2(M_n) \geq (24 - k)n - O(\log n)$.

(2) (Ceiling.) *With γ_h the archimedean growth rate of the reduced cross determinant h_n, one has the exact identity $\gamma_h = k^* \log 2 + H^*$. Consequently, for $k = k^* + \eta$, excluding the kernel output by a lower bound on the divisibility of h_n would require a divisor of h_n of size $e^{(\gamma_h + \eta \log 2/2)n}$, larger than h_n itself. No factorisation information about h_n can exclude it.*

(3) (Homogeneous congruences.) *Any further congruence imposed on $\mathcal{K}_n$ at a prime ℓ that is a unit for the rows caps the Smith invariant at $v_\ell(T_n)$; if the rows gain an extra factor of ℓ the kernel vector gains the same factor.*

(4) (Coefficient direction.) *If ℓ divides both coordinates of one row to higher order than the other, the kernel vector has $\ell \mid c_Z$; but the defining congruence $c_Z Y + c_N U \equiv 0 \pmod{S_n}$ already forces the same divisibility on every vector of $\mathcal{K}_n$ (net directional mass 0 for all $n \leq 60$), and the coefficient box does not shrink as $k \downarrow k^*$ ($\beta_Z(k^*) = E_N$).*

(5) (More rows.) *For three aligned rows the kernel vectors form a rank-two plane whose two short directions force the third successive minimum to expand by exactly the nominal gain.*

Parts (2) and (3) are due to GPT-5 ("Sol") in the project's research log; (1), (4), (5) were established in the audit documents `CATALAN_AUDIT`, `CATALAN_DIRECTIONAL`.

6.3. **Positivity.** Sign is the one datum valuations cannot see. Both source forms are strictly positive: $(-1)^m(Q_mG - P_m) > 0$ by Zudilin's double integral [31], and $J_n = 4B_nG - C_n > 0$ by Nesterenko's. Hence any coefficient vector in the closed positive cone (after orienting the Zudilin row) gives a nonvanishing form.

Proposition 6.3 (One kernel)**.** *With $w = xy/(1-xy)$ and $K_Z^{(m)}$ the Zudilin integrand of index m,*

$$K_N^{(n)}(x,y) = K_Z^{(3n)}(x,y)\, w(x,y)^n.$$

Hence for every (c_Z, c_N) the combined form is a single integral $\iint_{[0,1]^2} K_Z^{(3n)}\,(\alpha + \beta w^n)\,dx\,dy$.

Since w maps the open square onto $(0,\infty)$, the bracket has constant sign iff α, β do: Proposition 6.3 recovers the cone statement and nothing more. Moreover, for $P \in \mathbb{Z}[w]$ with nonnegative coefficients the form $\iint K_Z^{(3n)} P(w)$ is a positive combination of moments and is minimised at a vertex, so it never beats the best single moment; sums of squares $P = Q^2$ do cancel but are bounded below by $e^{-8.25n}$ independently of $\deg Q$, because the pushforward of $K_Z^{(3n)}$ to the w-line has a Pareto tail and all orthogonal-polynomial norms share one exponential order. An explicit positive construction in the style of Beukers therefore does not exist within this family.

Empirically the story is different from the pessimism above: for $n \le 44$ the shortest vector of $\mathcal{K}_n$ lies in the positive cone in 61% of cases, and the shortest vector *of the cone* has form decaying like $e^{-0.25n}$ at $n = 44$, tracking $\frac{1}{2n}\log\operatorname{covol}$. The rational control of Proposition 6.1 applies verbatim to these numbers.

6.4. **The remaining lemma.**

Lemma (P2)**.** *There is $\eta > 0$ such that for all large n the first minimum of $\mathcal{K}_n$, in the metric weighting the form by e^{-Fn}, is attained in the positive cone up to a factor $e^{o(n)}$.*

Proposition 6.4. *Lemma P2, together with the proved* 2*-adic package of* [12] *and the standard rates, implies that $G \notin \mathbb{Q}$. Conversely, Lemma P2 is false for every rational $G = a/b$: the kernel direction has scaled length $\asymp b\,e^{2Fn}$ and becomes the first minimum for $n \gtrsim \log b/|F|$.*

Lemma P2 is thus not a lattice-theoretic lemma but a restatement of the irrationality of G: any proof must distinguish G from G^* at precision growing with n, i.e. must use analytic information about G. What the two-row method contributes is an exact reduction with no loss—worthiness 1 is proved [12], and irrationality is equivalent to the single statement P2 about one explicit moment lattice. We return to this in Problem 7.5.

7. Problems

Problem 7.1 (Rigidity)**.** Prove Conjecture D for Zagier's rows B, C, F and Cooper's s_{18} at $p = 3$, and for Domb and $(12,4,16)$ at $p = 2$: identify the common p-adic limit with a p-adic L-value via the p-adic Eichler integral of the Eisenstein source (Calegari's method at level 6, 8, 9, 12). The explicit sources and levels are known (Section 2; $(12,4,16)$ lives on $\Gamma_0(8)$ with $t = (\eta(\tau)\eta(8\tau)/\eta(2\tau)\eta(4\tau))^8$).

Problem 7.2 (Scalar barrier)**.** Prove that for trivial character, $w \ge 3$, and Γ genus zero commensurable with $SL_2(\mathbb{Z})$, every integral Hauptmodul system of Sym^w type has $\log|t_2| - (w+1) < 0$. The census of [9] ($N \le 25$, eta-quotient Hauptmoduls) supports this; a proof would formally close the scalar route to $\zeta(5)$.

Problem 7.3 (Decaying partners with $\kappa_p > 0$)**.** By Proposition C and the design rule of Section 5, the only way to raise the quality of a two-row construction is a decaying row with p-power denominators. Find hypergeometric (Padé or Rhin–Viola type) rows for $L(2, \chi_{-3})$ at $p = 3$, for

$L(f,2)$ with $f=\eta(2\tau)^3\eta(6\tau)^3$ at $p=2$ (engine: $\sigma_2=2$, $k=2$, needs $\Lambda_{\mathrm{dec}}>14.8$ for $\sigma_{\mathrm{dec}}=1$, >29.6 for $\sigma_{\mathrm{dec}}=2$), and for $\zeta(3)$ at $p=2$ (engine: Domb). For $L(2,\chi_{-3})$ four aligned rows exist and none decays.

Problem 7.4 (New sporadic sequences)**.** Enumerate (Γ,t,f) on genus-zero $X_0(N)$ with t an eta quotient and f selected by the growth filter (coefficients of $\sum a_n t^n$ bounded polynomially), rather than by an eta-quotient ansatz for f. A $5\cdot 10^5$-pair scan recovered all known sporadic rows and the $L(f,2)$ row; the modular forms f of the known rows are Eisenstein combinations, not eta quotients, which earlier searches wrongly excluded.

Problem 7.5 (Adelic Padé)**.** Replace lattice selection in the Catalan construction by an explicit construction with a determinant-type nonvanishing argument. Two facts constrain the problem. First, for the moment family $M_{n,j}=\iint K_Z^{(3n)}w^j=A_{n,j}G-B_{n,j}$ one has the exact 2-adic closed forms (verified for all $j\le 3n$, $n\le 5$)

$$v_2(A_{n,j})=3-2(6n+j)+s_2(3n)+s_2(3n+j),\qquad v_2\Big(G_2-\frac{B_{n,j}}{A_{n,j}}\Big)=24n+4j-1-2s_2(3n)-2s_2(3n+j),$$

which contain the two 2-adic theorems of [13] as the cases $j=0,n$; the cross valuation of any two moments is $2v_2(d)+5-|v_2(A_{j_0})-v_2(A_{j_1})|$ exactly, so the “$24n$ bridge” is an ultrametric consequence of the denominators alone. Second, the 2-adic rate of these rows is the archimedean order of vanishing converted by Legendre’s formula on the beta prefactor: there is no independent 2-adic contact order in this family, hence no two-place Hermite–Padé problem with separately tunable conditions. The useful nonvanishing criterion is *Lemma AP*: if $\mu(e)=v_2(G_2-B_e/A_e)\to\infty$ along a sequence of outputs, then $A_eG-B_e\neq 0$ eventually (using only $G_2\notin\mathbb{Q}$); kernel vectors are exactly those with bounded μ, and every short vector tested has $\mu\approx(24-k)n$. The problem is therefore to construct, for a function taking the values G and G_2 at an archimedean and a 2-adic point, a p-adic Padé system in Beukers’ sense whose 2-adic contact order is *independent* of its archimedean one, so that $\mu\to\infty$ is built in and the archimedean size is an explicit remainder estimate. Within the Zudilin–Nesterenko family the residual statement P2′ (first minimum outside the thin sublattice of bounded μ) is weaker than P2 but still of irrationality strength.

Problem 7.6 (Free integrations)**.** Characterise, in terms of the source Φ, when $D^{-1}\Phi$ is already integral in the t-lattice (Cooper’s $(n+1)\mid A_n$, proved by Bogner). Each free integration lowers k by one, which is worth more than any p-adic device.

Appendix A. Verification scripts

All computations are exact (PARI/GP, rational arithmetic) unless stated; each script prints what it checks. Repository paths are relative to the project root.

`verify/period_annihilation.gp`: Dirichlet-polynomial facts of Section 2: Beukers’ annihilator; level-8 and level-16 Gaussian-binomial projectors; the level-12 $\zeta(7)$ parent kills $s=0,2,4,6,8$ with target $\frac{209}{1728}\zeta(7)$; $\beta(4)$ inner/outer vectors; the χ_{-3} purified vector. 12 PASS.

`verify/beta4_lambert.py`: The $\beta(4)$ Lambert identity at $\tau=i/\sqrt{24}$ to 40 digits (mpmath).

`lattice/zagier_padic.gp`, `zagier_limits.gp`, `zagier_CF.gp`: Zagier A–F: $v_p(a_n)$ bounds, slopes, archimedean limits by `lindep`, 3-adic coincidence of B, C, F with factors $\frac12,\frac12,\frac58$.

`lattice/third_order.gp`, `zeta3_align.gp`: Third-order sporadics; Domb vs $(12,4,16)$ 2-adic alignment with factor $\frac43$, $n\le 400$.

`lattice/census/`: Cooper rows (corrected initial conditions), $\eta=(11,5,125)$, $\delta=(7,3,81)$, level-24 $\zeta(7)$ (A_n, B_n, d_n^7, limit).

`lattice/zeta3_lattice/`: Integer model of Domb and $(12,4,16)$, exact 2-adic tails, roof lemma, the master-formula evaluation $\delta=0.9009531687$, LLL harness, the $\Gamma_0(8)$ parametrisation of $(12,4,16)$.

`lattice/catalan_audit/`: Exact Zudilin and Nesterenko $(4,7)$ rows, agreement with the project CSVs, the rational-G^* control experiment, lattice index by explicit basis.

`lattice/catalan_directional/`, `catalan_positivity/`, `catalan_explicit/`: The absorption, positivity and explicit-moment computations of Section 6; the one-kernel identity $K_N^{(n)} = K_Z^{(3n)} w^n$ to 57 digits.

`lattice/paper_tables/`: Generators for Tables 2 and **??**.

The Lean formalisation of the $1-\varepsilon$ theorem is the project `lean-catalan-worthiness`; `AXIOM_CHECK.lean` confirms that `catalan_worthiness_one_sub_eps_nesterenko` depends only on `propext`, `Classical.choice`, `Quot.sound`.

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